

Roehampton Library Retrofit - Technical Report

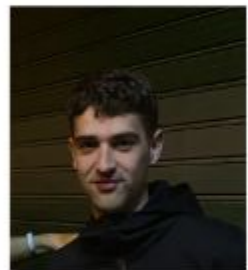
Group 2



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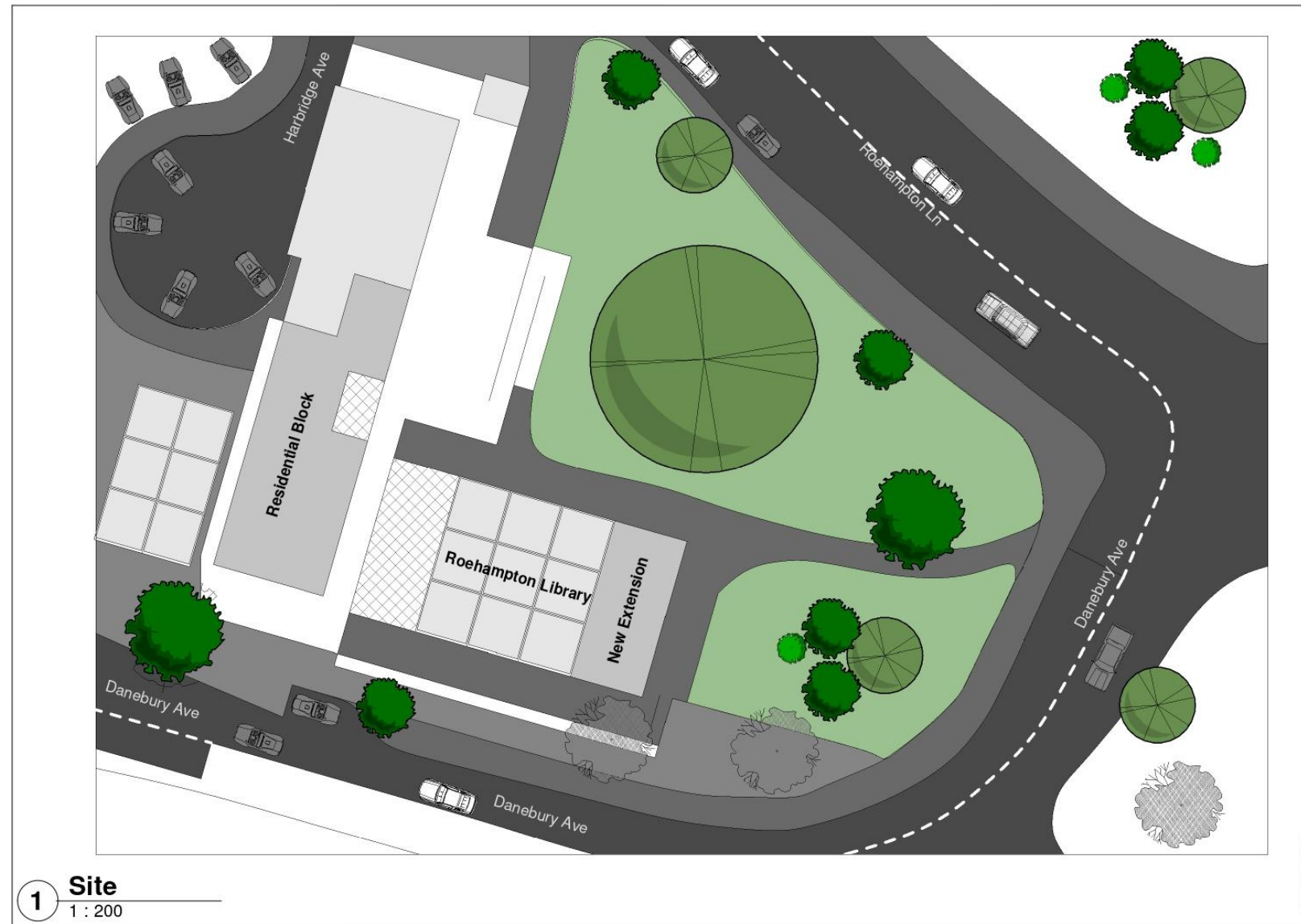


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Site Plan (1:200) and Building Assessment



Current Building condition

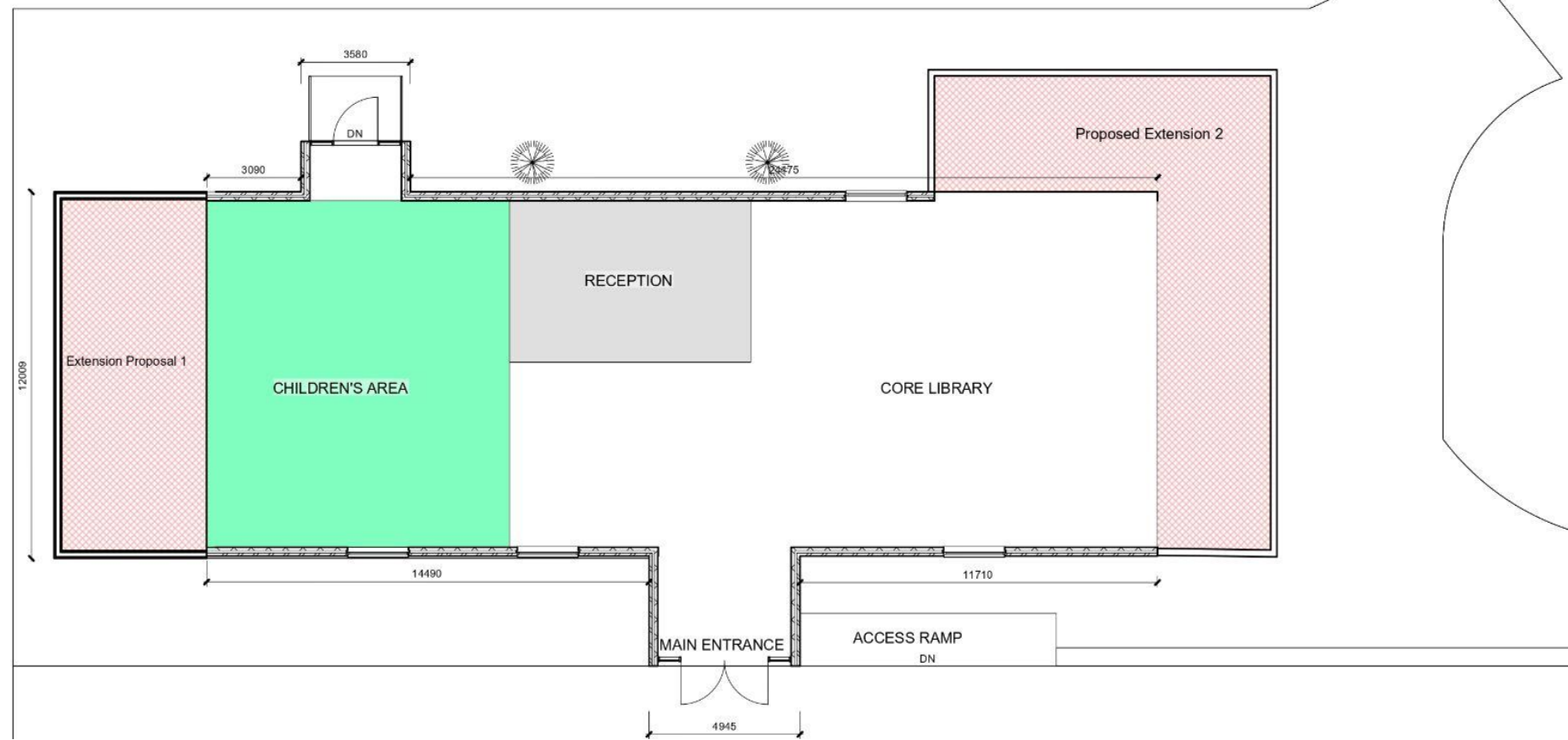
- Poor energy efficiency, significant heat loss through single glazed windows, roof and existing walls
- Gas boiler and radiators causing high maintenance costs and a reliance on mains gas (amidst national move towards net zero)
- Unknown condition of existing concrete pillar and beam frame
- Poor indoor air quality, arising from current ventilation systems
- Lack of space, requiring a 15-20% extension of the floor area
- Outdated Fire safety and Accessibility

Key Extension Issues

- Albrook House residential building above prevents any vertical expansion
- Lateral extension from the eastern wall is the most feasible, due to the residential car park and significant level change (to the west), Danebury avenue (to the south) and Residents access and ramp (to the north)

Extension Feasibility Studies

Floor plan with 2 proposed Extensions



Extension Proposal 1

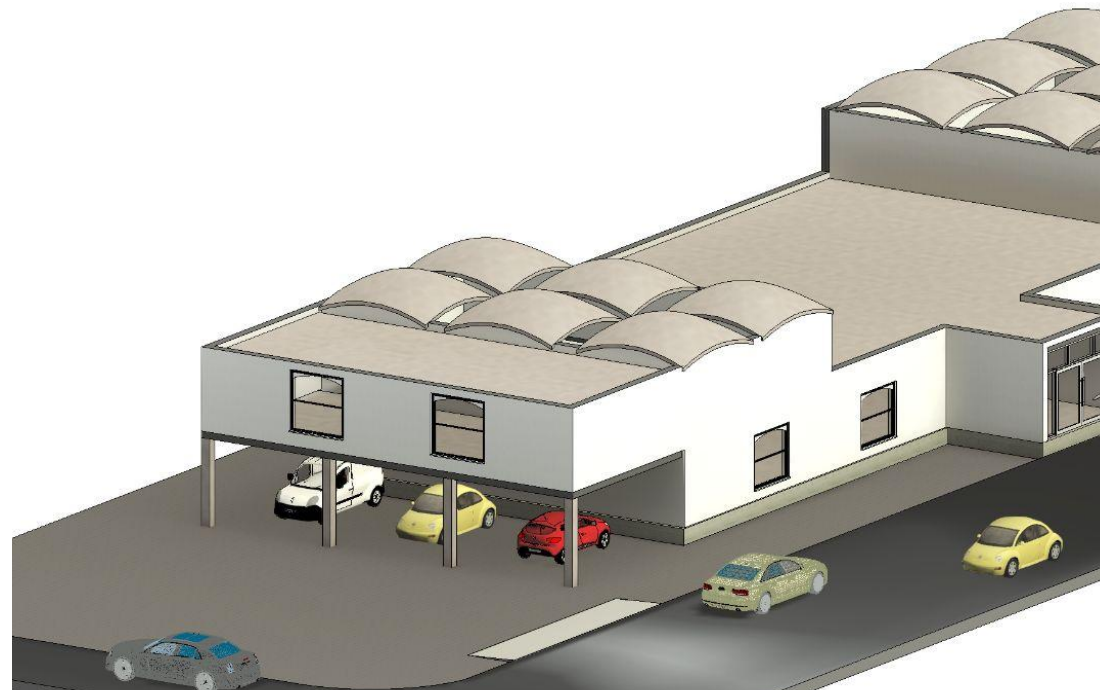
- Extending by 15% from the western wall of the library where the children's area was located
- This would have required support columns and a subfloor and would be much more complex due to the significant level change to the Albrook House car park below. Increasing time and cost

Extension Proposal 2

- Achieving a 15% extension through both the North and Eastern Wall
- Would have required demolition of 2 walls instead of 1, as well as increasing the external works as the residents access path would need to be moved on the north side. Increasing time, cost and disruption to residents

Figures 1,2:

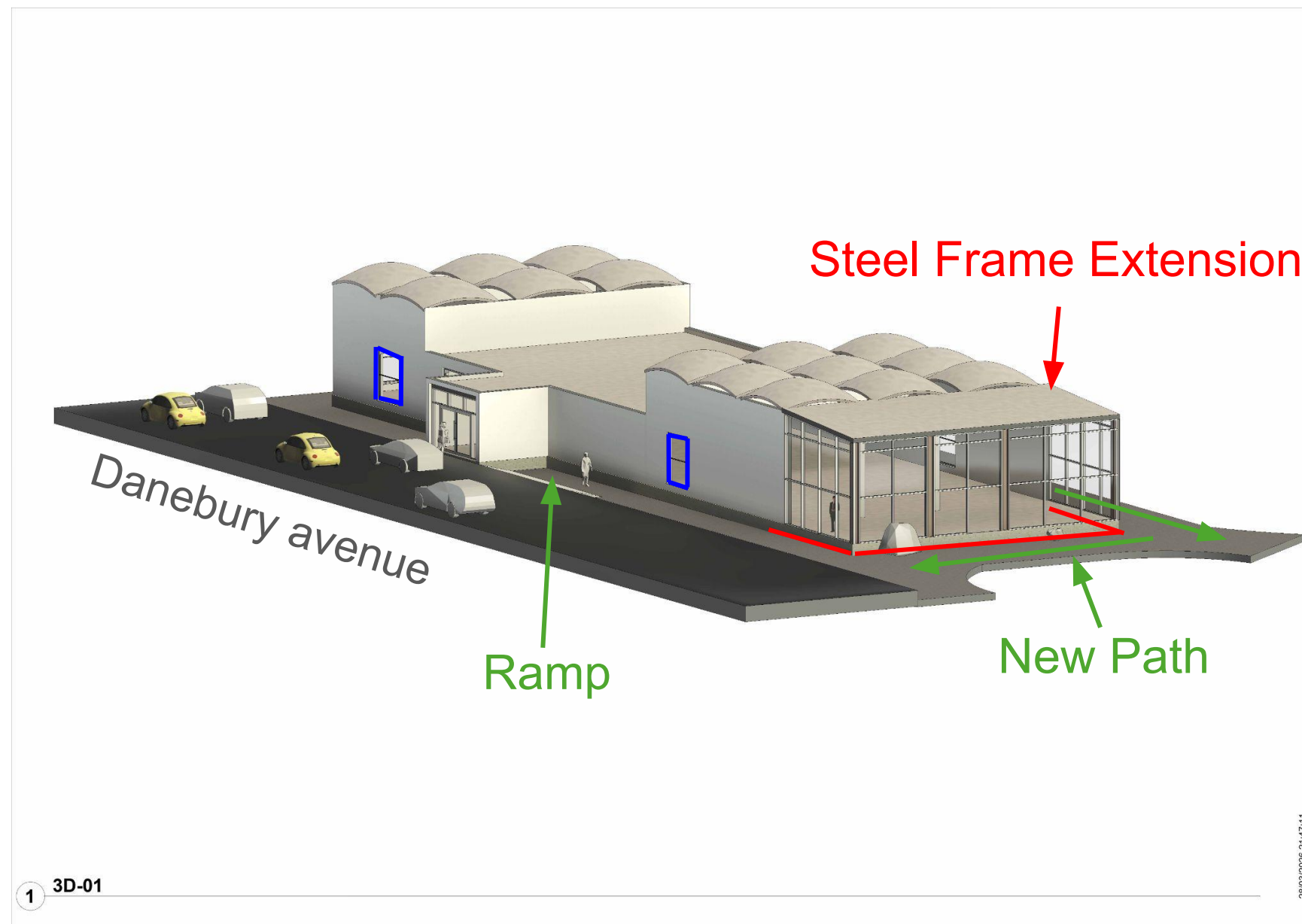
A floor plan showing 2 proposed extensions during feasibility studies, before deciding on the final design. As well as a concept model of the Extension proposal 1, over the car park



Proposal: Deep retrofit with Extension - 5 Phases

Phase 1 - Extension

- Assembly of a steel framed extension, with double glazed panelling
- Residential access path must be redirected
- Library can remain partially open during initial extension works



Phase 5 - Library Reopening

- Library retains original character with all needed improvements met

Phase 4 - Fabric upgrade

- Replacement of large window panels with **Double Glazing**

Phase 2 - External works

- The Residents access path must be redirected around the new extension, maintaining access from Roehampton Lane, to Danebury Avenue
- Construction of a ramp on Danebury avenue side next to the stairs for improved accessibility

Phase 3 - Internal works and fit out

- Removal of all library items which are sent to a storage facility.
- Stripping of gas boiler, radiators, old lighting systems
- Installation of **LED lighting, MVHR system, Air Source Heat Pump and Underfloor Heating**

Building Regulations and PAS 2028 Compliance

Building Regulations (UK)

- **Part L - conservation of Fuel & Power**

The replacement of single-glazed with double-glazed windows reduces the window U-Value from 5.0 W/m²K to 1.4 W/m²K, significantly reducing heat loss, hence conserving energy.

- **Part F - Ventilation**

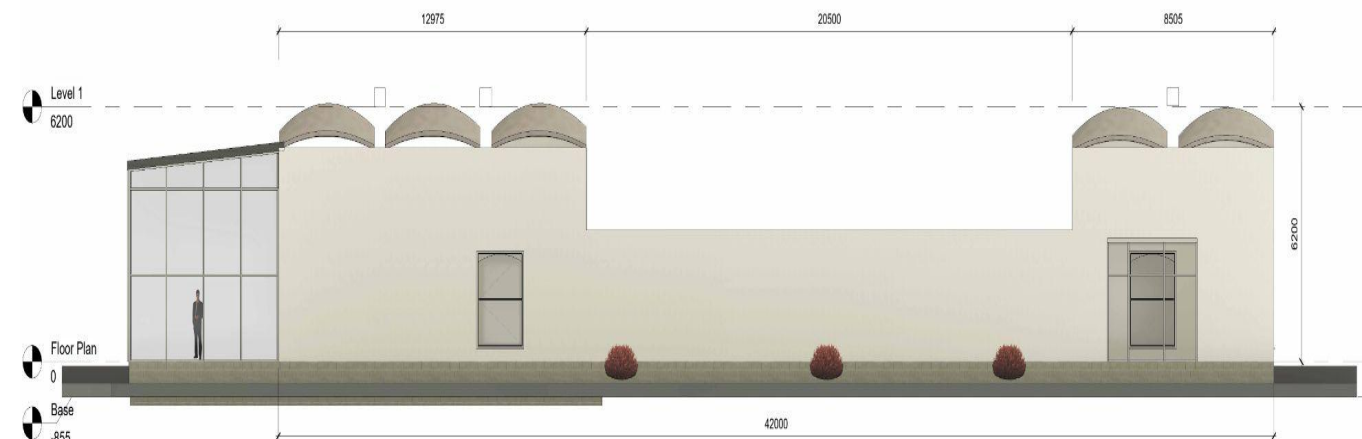
MVHR systems significantly help to improve indoor air quality and regulate overall airflow. Directly addressing a key problem in the building.

- **Part M - Accessibility**

The new and inclusive design with careful considerations of the doors and corridors width, circulation space, and access ramps helps make the building accessible and functional.



East Elevation



North Elevation

PAS 2028 Retrofit Principles

- **Whole Building Approach**

Proper integration of building fabric, ventilation and heating systems, ensures a coordinated and efficient retrofit solution.

- **Energy Performance Solution**

This reduces heat loss and carbon emissions through a fabric and a services upgrade from double glazed windows, and natural lighting improvement via the extension. It enhances overall energy efficiency.

- **Occupant Health and Comfort**

Occupant comfort is achieved through Improved internal air quality through controlled ventilation, stabilized internal temperatures, and increased daylighting.

Floor Plan (1:100)



Extension Area

- Achieves **15%** increase of Floor Area, creating a larger core library space
- Eastern wall will be demolished during Phase 3 (Internal Works), however concrete pillars will remain due to structural integrity
- Double doors create a new fire exit onto the nearby grass area, improving the fire safety plan of the building

Accessibility

- A new access ramp for wheelchair users on the Danebury Avenue side of the library, as part of Phase 2 (External Works)

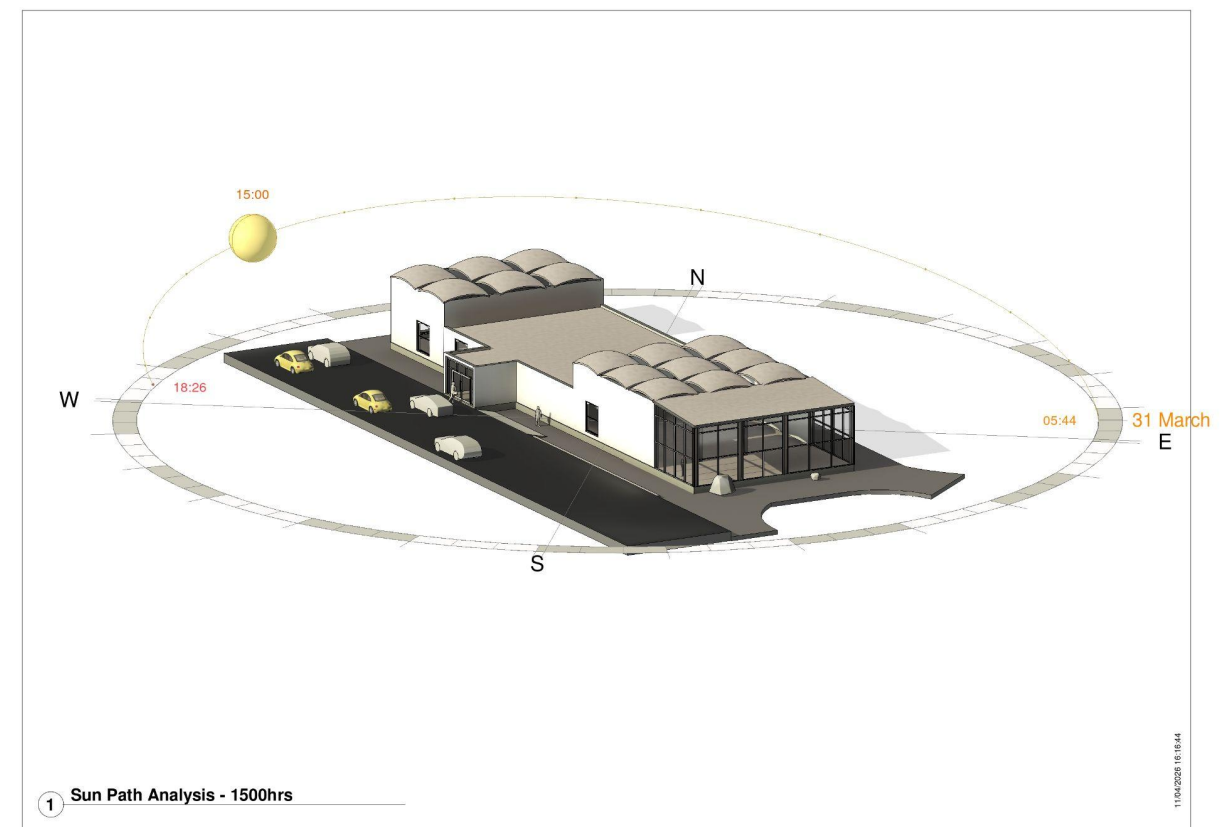
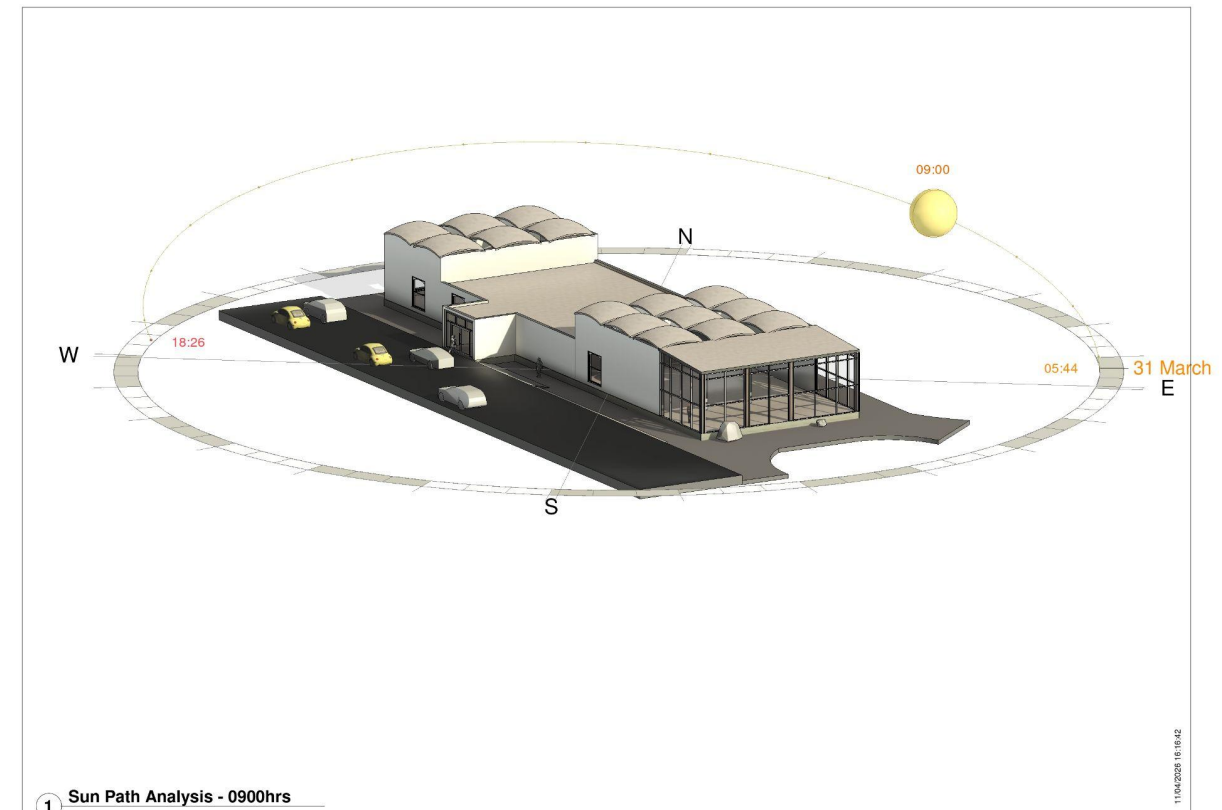
Passive Design Benefits

- The double glazed panelling will allow for much more natural daylight to enter the library during opening hours. Reducing the burden on electrical lighting

Sun Path Analysis

- The South east facing extension will provide abundant natural daylight throughout the library's operating hours. The double glazed facade allows natural daylight to come through
- Oriented southeast, which is optimal in the UK. It maximizes passive solar gain, thereby enhancing energy efficiency and supporting overall sustainability objectives.

Library open hours: 10-6pm.



Fabric of external wall

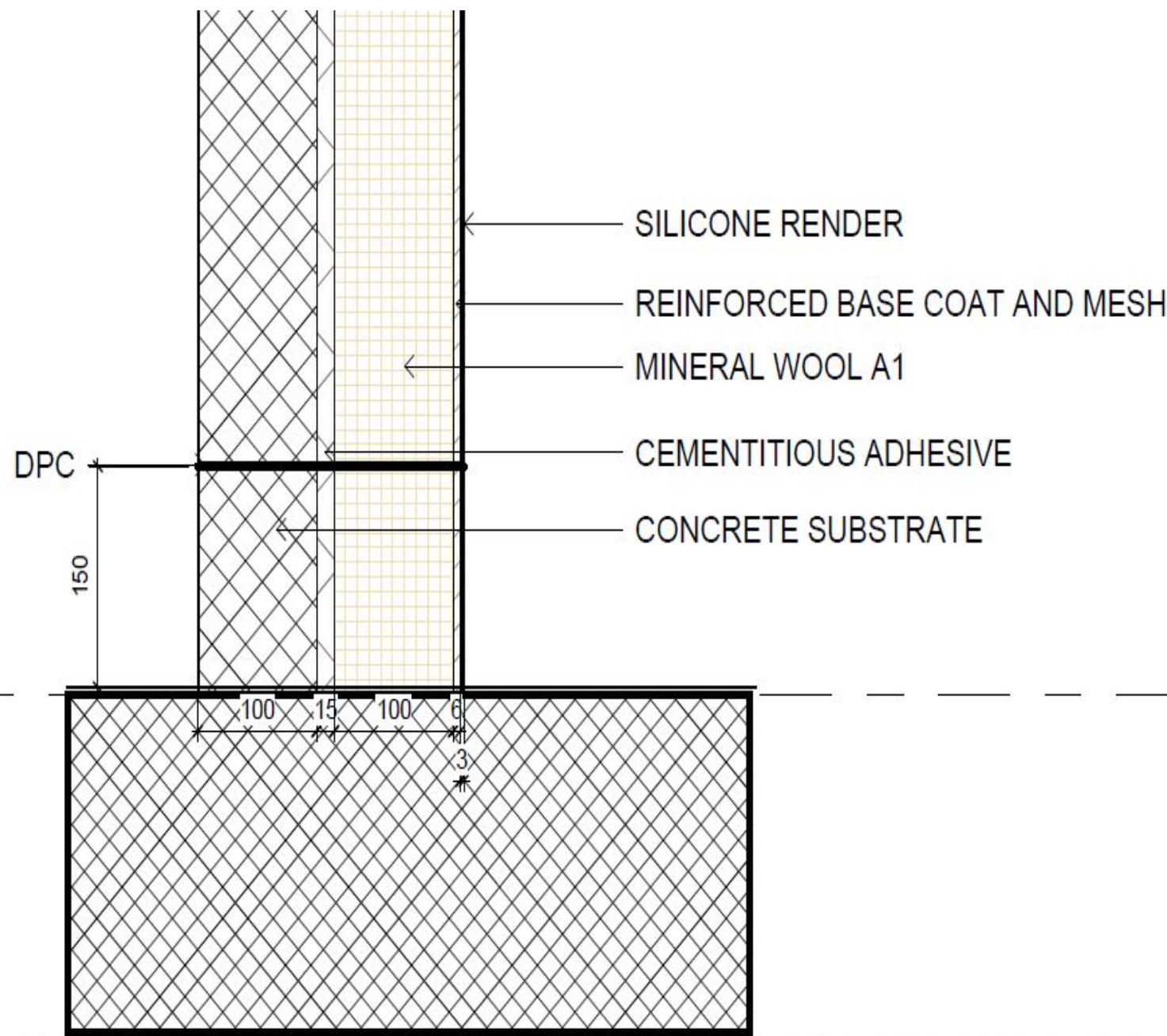
Render repels rain still letting the wall breathe, preventing damp and mould inside

Mineral wool A1 is fire safe as it is non combustible.

Insulation achieves U- value of 0.30 w/m²k , reducing heat loss from external walls

Start positioned 150mm above dpc stops heat escaping at vulnerable wall to ground junction

This system covers meets the uk regulation plus PAS 2035/2030, so the building is compliant for retrofit work.

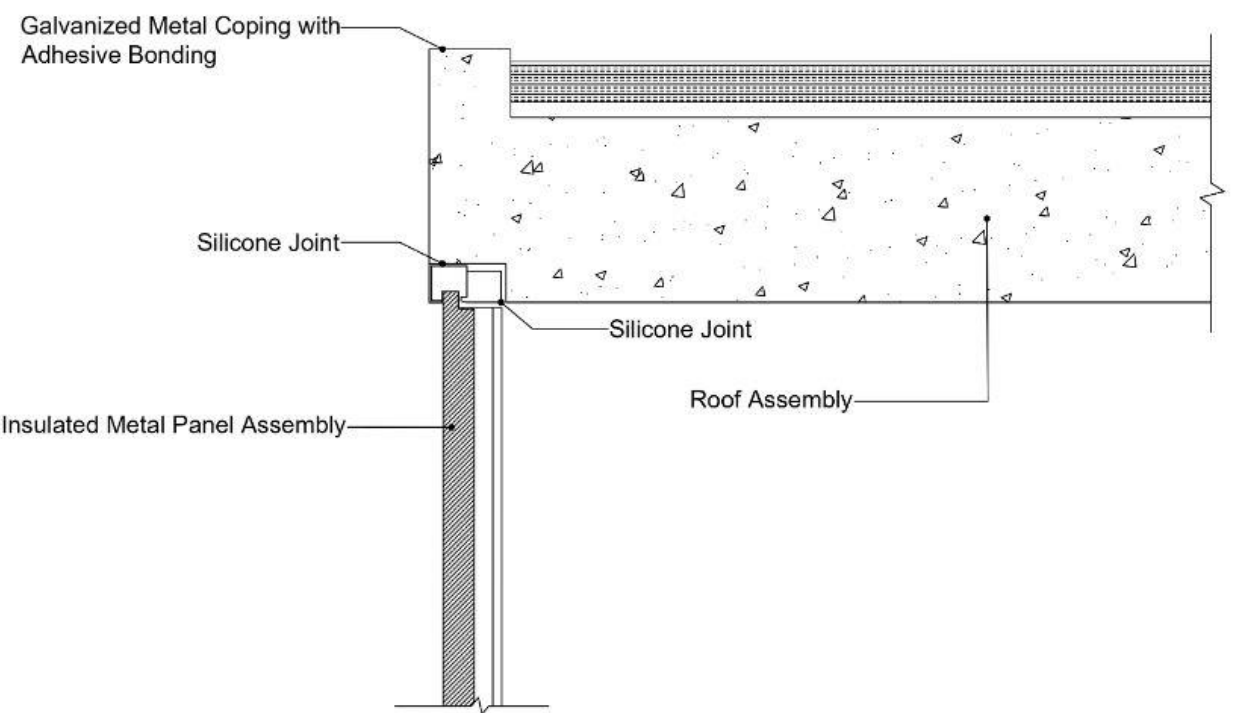


External wall insulation -material specification table

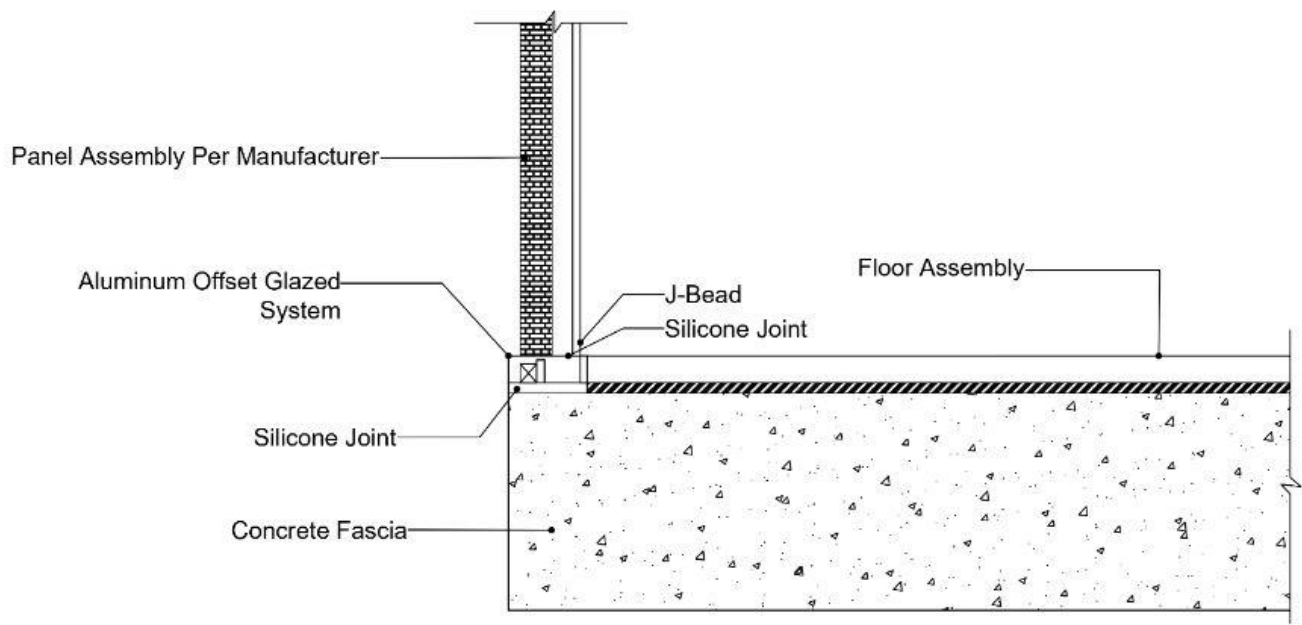
No.	LAYER	MANUFACTURE	PRODUCT	TECHNICAL FUNCTION	REGUALTION ALINGMENT
1	SILICONE RENDER FINISH	Weber (Saint-Gobain) uk.weber/ewi	webersil TF Silicone Textured Finish	Provides weather protection and decorative finish. Hydrophobic surface repels driving rain while remaining vapour permeable, allowing the wall to breathe.	Meets Part C by preventing moisture ingress. Complies with Part B fire requirements rated B-s2,d0. BBA certified under certificate no. 91/2691.
2	REINFORCED BASECOAT AAND MESH	Wetherby Wall Systems wall-systems.co.uk	Heck K+A Basecoat + Fibreglass Mesh 160g/m²	Polymer-modified cementitious render applied at 4–6mm with alkali-resistant fibreglass mesh fully embedded. Provides structural reinforcement and impact resistance over the insulation.	Satisfies Part A by reinforcing the insulation layer against structural impact. Meets Part C moisture resistance requirements. Fully compliant with PAS 2035:2023 retrofit standard.
3	Mineral Wool A1 Insulation	ROCKWOOL Ltd rockwool.com/uk	External Wall Dual Density Slab $\lambda = 0.038 \text{ W/mK}$ Euroclass A1	Non-combustible stone wool insulation board. Dual density construction — soft face accommodates substrate irregularities, hard face provides stable render surface. Minimum 7 mechanical fixings per m² required.	Achieves Euroclass A1 to BS EN 13501-1 — the highest non-combustible rating, satisfying Part B fire safety for buildings of any height. Delivers U-value of $0.30 \text{ W/m}^2\text{K}$ meeting Part L 2021 energy efficiency target. Breathable structure satisfies Part C

4	Cementitious Adhesive	Wetherby Wall Systems wall-systems.co.uk	Wetherby Bedding Adhesive Min. 40% coverage	Polymer-modified cementitious adhesive. Applied using perimeter bead and 3 centre dabs method. Bonds insulation to concrete and accommodates minor substrate irregularities up to 10mm.	Meets Part A fixing requirements by securing insulation to substrate. Satisfies Part C by sealing substrate against moisture. Must be installed by PAS 2030:2023 certified installer.
5	Concrete Substrate	Existing structure Retrofit substrate	Cast In-Situ Concrete Pull-out tests required before installation	Existing concrete wall forming the structural base for the EWI system. Surface must be clean, sound and free from contaminants. Pull-out tests confirm fixing capacity before works begin.	Satisfies Part A through pull-out testing of substrate. PAS 2035 requires a pre-installation whole-house survey to assess existing fabric condition before EWI is specified.
6	Aluminium Starter Track	Wetherby / Weber Base of system	Aluminium Base Rail Fixed above DPC @300mm centres	Mechanically fixed aluminium rail supporting insulation boards at the base of the system. Prevents downward movement. Set min. 150mm above finished ground level, fixed above DPC.	Satisfies Part A mechanical support. Meets Part C by terminating correctly above DPC. PAS 2035 requires EWI to extend min. 150mm below DPC to eliminate cold bridging at base junction.

Door Header and Sill Details



A DOOR HEADER DETAILS
01 SCALE 1:10



A DOOR SILL DETAILS
01 SCALE 1:10

Refer to Section 1/B on Floor Plan Scale: 1:10

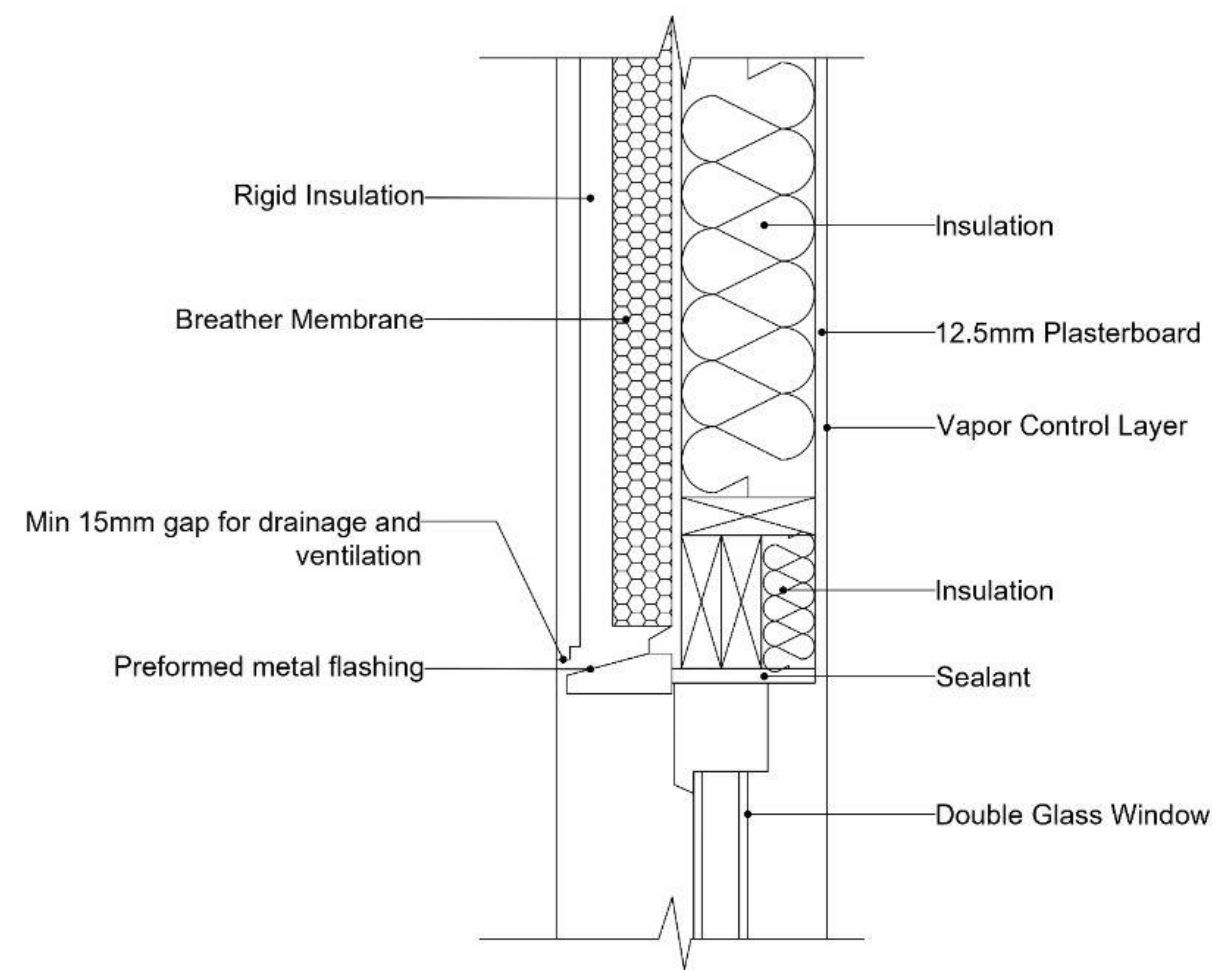
Door Header

1	Galvanized Metal Coping with Adhesive Bonding	Weatherproof top cap that protects the assembly from rain and gives a clean finished edge.
2	Silicone Joint	Flexible sealant that prevents water and air leakage while allowing for thermal movement.
3	Insulated Metal Panel Assembly	Provides structural support, thermal insulation, and finished surfaces in one system.

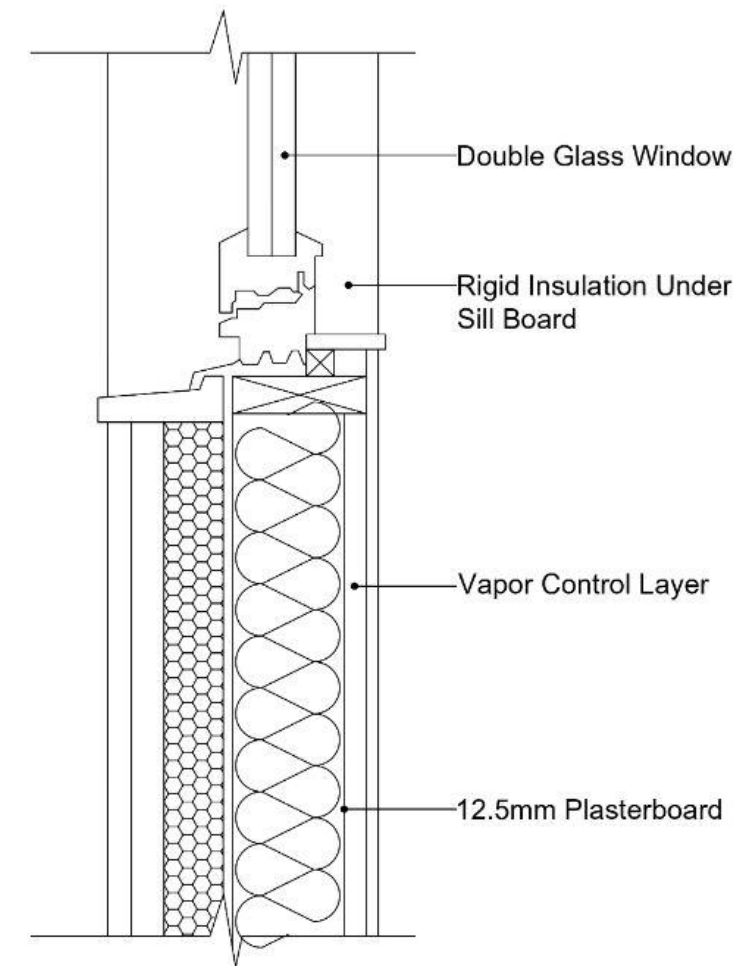
Door Sill

1	J-Bead Silicone Joint	Provides a clean and durable termination detail for the silicone sealant at the sill.
2	Aluminum (Offset Glazed) System	Aluminum framing that supports glazing. The offset design aids drainage and improves performance.
3	Floor Assembly	The structural floor below the door threshold ensures proper load transfer and level transition for easy accessibility.

Window Header and Sill Details



A WINDOW HEAD DETAIL
01 SCALE 1:10



B WINDOW SILL DETAIL
02 SCALE 1:10

Refer to Section 1/C on Floor Plan Scale: 1:10

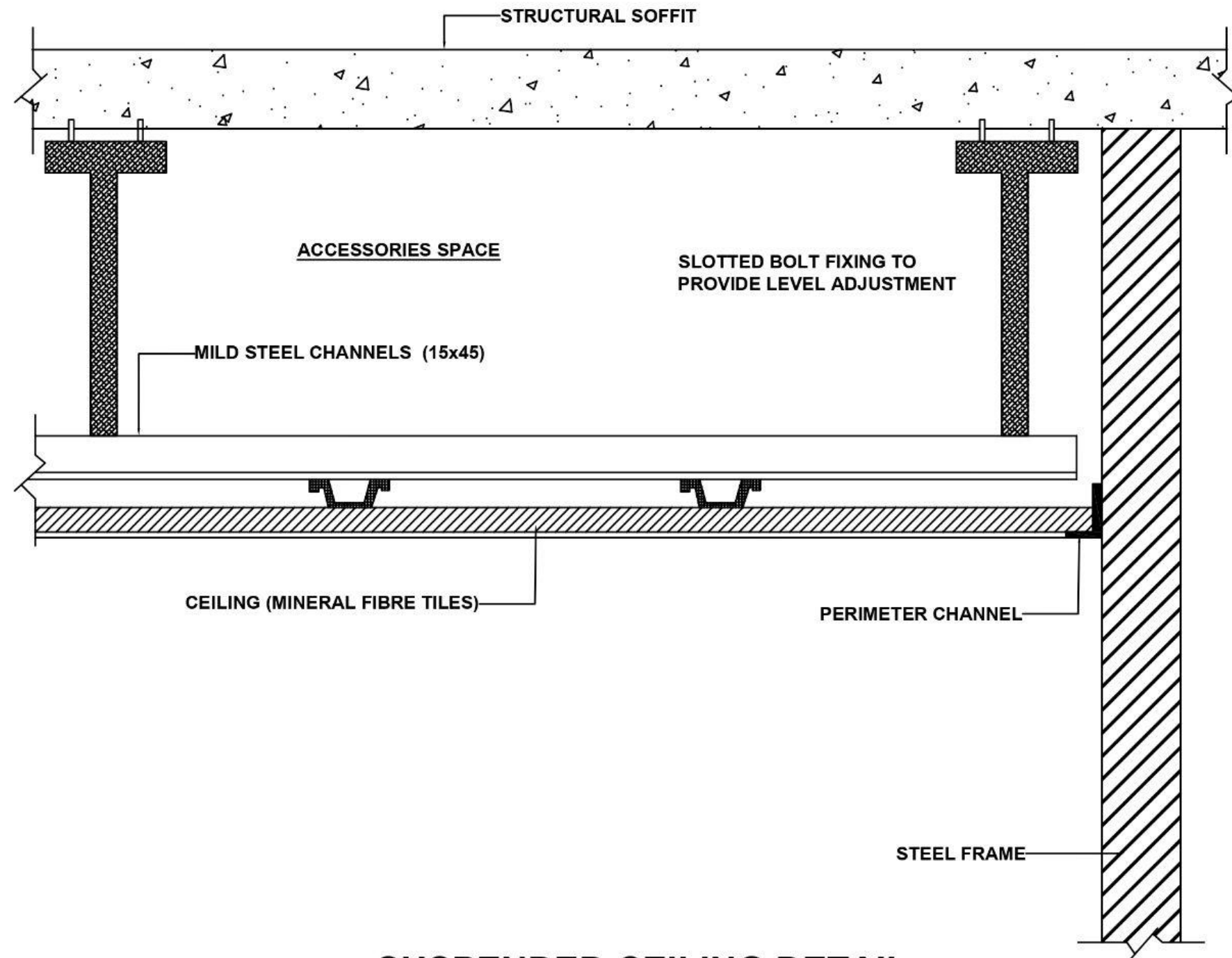
Window Head Detail

1	Rigid Insulation	Provides high-performance external insulation, maintaining continuity around the window opening to minimise thermal bridging.
2	Breather Membrane	Allows vapour to escape outward while protecting insulation from wind-driven rain.
3	Vapour Control Layer (VCL)	Prevents moisture migration from the interior into the wall build-up.
4	Preformed Metal Flashing	Directs water away from the window head and creates a weatherproof transition.
5	15mm Drained & Ventilated Gap	Allows drainage and ventilation of any moisture that enters the cavity, preventing trapped moisture.
6	Sealant	Creates an airtight and weathertight seal between the window frame and surrounding construction.
7	Double Glazed Window	High-performance glazing unit contributing to improved thermal and acoustic performance.

Window Sill Detail

1	Double Glazed Window	This is the primary glazing element with low U-value for energy efficiency.
2	Rigid Insulation Under Sill Board	Maintains thermal continuity beneath the sill, reducing cold bridging at the base of the window.
3	Vapour Control Layer (VCL)	Prevents moisture migration from the interior into the wall build-up.
4	Insulation	Continues the wall insulation layer for consistent thermal performance.

Suspended Ceiling Detail



SUSPENDED CEILING DETAIL

Refer to Section 1/D on East Elevation Scale: 1:10

SUSPENDED CEILING DETAIL

1	Structural Soffit	Provides the primary structural support from the existing concrete frame or new steel structure above.
2	Mild Steel Channels (16x49)	Main support grid for the suspended ceiling system. Offers strength while keeping the system lightweight.
3	Slotted Bolt Fixing	Allows precise level adjustment during installation and accommodates structural deflection or movement.
4	Perimeter Channel	Creates a clean, secure edge detail around the room perimeter and provides additional support for ceiling tiles.
5	Mineral Fibre Ceiling Tiles	Acoustic and aesthetic finish. High sound absorption (NRC up to 0.90) improves internal acoustics for a quiet library environment. Also contributes to fire resistance.
6	Accessories Space	200–300mm clear void above the ceiling for services (lighting, sprinklers, HVAC, cabling, and fire detection). Allows easy future maintenance and flexibility.
7	Steel Frame Integration	Connects the suspended ceiling to the main steel extension structure, ensuring overall stability and coordination with the lightweight roof system.

ROOF THERMAL PERFORMANCE CONTRIBUTION

- **Thermal Buffer & Insulation Continuity:** The suspended ceiling creates a service void that acts as an additional insulated plenum. This allows extra insulation to be installed above the ceiling tiles, enhancing the roof's overall U-value and reducing heat loss and gain through the roof.
- **Improved Energy Efficiency:** Together with the roof insulation above, this detail helps achieve a roof U-value of $\leq 0.15 \text{ W/m}^2\text{K}$, contributing to Part L compliance and lowering the library's heating and cooling demand.
- **Condensation Control:** The ventilated void and breathable mineral fibre tiles help manage humidity and reduce the risk of condensation on the underside of the roof structure.

Thermal Bridging Mitigation at Windows and Door Openings

- **Continuous Insulation** - The details maintain uninterrupted insulation from the roof assembly through the insulated metal panels down to the floor. This eliminates major thermal breaks at the critical junctions, significantly reducing heat gain.
- **Thermal Breaks in Aluminum** - The offset glazed aluminum system incorporates thermal breaks that separate the exterior and interior aluminum profiles. This reduces conductivity by up to 70–85%, preventing the aluminum frame from acting as a major heat transfer path.
- **Airtightness** - Multiple high-quality silicone joints create an airtight seal. This is critical because air infiltration can account for 30–50% of total heat loss or heat gain in poorly sealed buildings. The flexible silicone also accommodates thermal movement without cracking.

Thermal Performance Improvement
(Existing vs Proposed U-Values)

	Element	Existing	Proposed
1	Wall	1.5	0.30
2	Windows & Doors	5.0	1.4
3	Roof	1.0	0.18

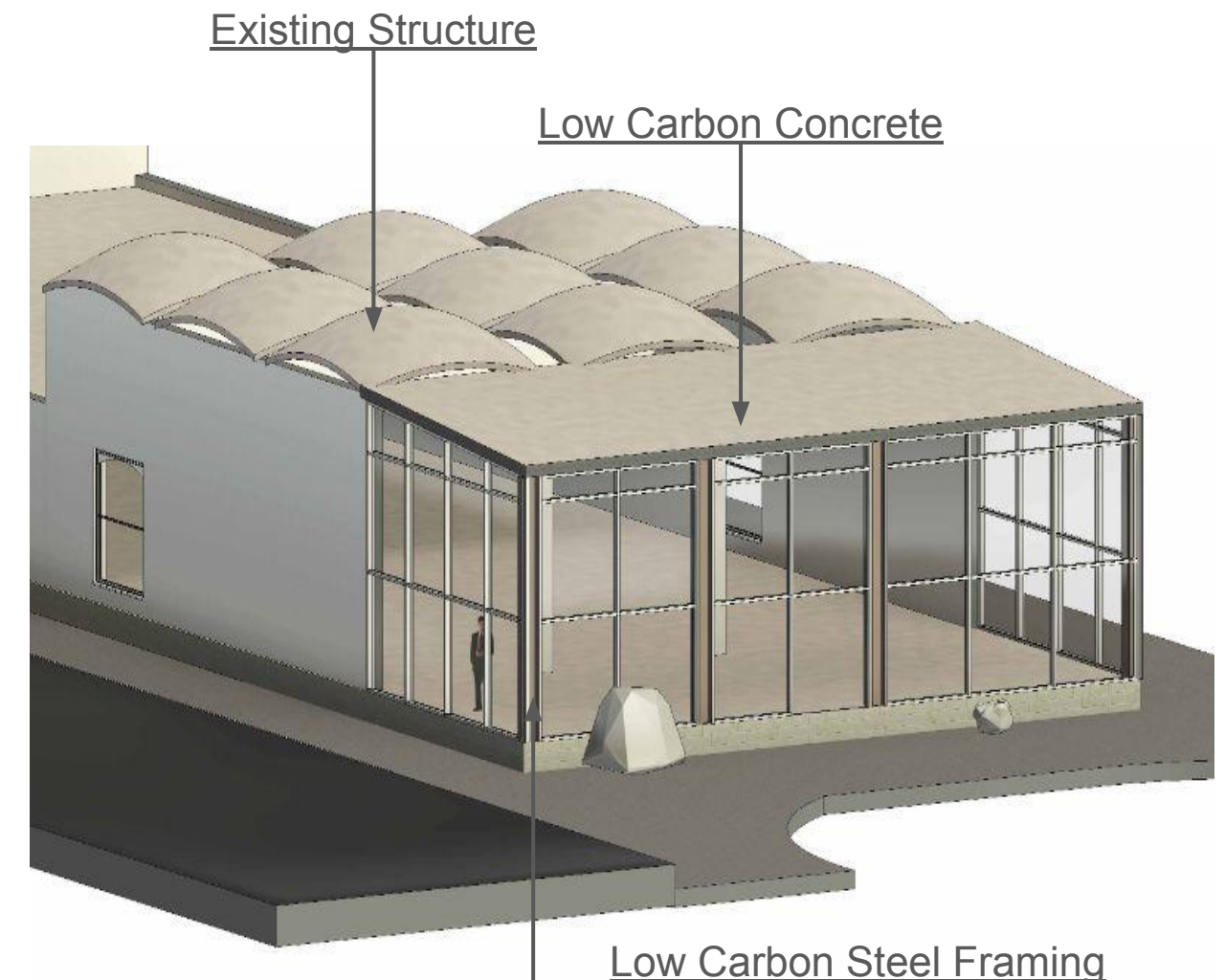
Airtightness & Moisture Management Strategy

- **Airtightness** - A continuous airtight line is maintained on the inner face of the building envelope using airtight tapes, membranes, and silicone joints around all windows, doors, and service penetrations. This targets an air permeability of $\leq 5 \text{ m}^3/\text{h}/\text{m}^2$ @ 50Pa, significantly better than minimum requirements.
- **Thermal Breaks in Aluminum** - A robust vapour control strategy is implemented with a high-performance VCL on the warm side of the insulation. The offset glazed aluminium system and J-bead details provide drainage channels for any incidental moisture. This combination prevents surface and interstitial condensation, protecting the existing concrete frame and new timber/steel elements in London's humid climate.

Material Selection Strategy

The material palette prioritises reuse, low carbon, and durability:

- **Reuse of existing concrete frame:** This significantly reduces embodied carbon by avoiding demolition and new foundations.
- **Low-carbon materials:** responsibly sourced low carbon concrete and steel, recycled aluminium glazing systems, and mineral wool insulation with low Global Warming Potential.
- **Durability:** Long-life materials (expected 50+ years) reduce future maintenance and replacement impacts.



Refer to 3D-01

All selections support circular economy principles and align with the project's sustainability targets.

Overall Extension Design Justification

The proposed east-side extension uses a lightweight steel frame with high-performance glazing. This design was chosen for the following reasons:

- **Lightweight steel frame** enables fast, low-disruption construction on a constrained urban site with limited access and neighbouring residential properties.
- **Extensive glazing** maximises natural daylight into the new reading and community spaces while reducing artificial lighting demand
- **Southeast-side location** respects site constraints: it avoids overshadowing neighbouring properties to the west, utilises the existing main entrance for improved flow, and allows construction without closing the library for extended periods.

Overall, the design improves buildability, minimises structural loading on the retained concrete frame, and enhances the user experience through better daylight.

Sustainability in the Construction Programme

Creating an Energy efficient Building

- Through removing mains gas and installing an air source heat pump and underfloor heating system. The libraries central heating will be fossil fuel free.
- The double glazing will reduce thermal heat loss from within the building. Placing less demand on heating systems.
- The double glazed extension will provide lots of natural daylight and passive solar gain during open hours



Low Carbon construction methods

- Prefabricated steel not only saves time on site, but reduces any waste from cutting/altering and any potential damage from long term storage

Recycling of old Building fabric

- Careful removal of the old window panelling can allow the glass to be recycled.
- Similarly all internal equipment such as radiators, the gas boiler and material from the eastern wall demolition can be recycled following their removal

Indicative Cost Plan

Column 1	Constituent	Cost (£)	Total (£)	Pricing Source
1	Facilitating Works -No major demolition needed -Groundworks in removing existing concrete path (£200p/m2), Area of Concrete path (19.2m2) -Skip Hire (£200 average fee)	200 p/m2 +£200 skip fee	£4440	Checkatrade https://www.checkatrade.com/blog/cost-guides/concrete-removal-cost/
2.1	Extension and Fabric (Phase 1 + 4 works) -Steel frame extension, substructure, glazing and subfloor, eastern wall demolition (£2000p/m2) -Double glazing (2x£2500)	£2000p/m2 +£5000	£181,000	https://www.fmb.org.uk/resource/single-storey-extension-costs-a-practical-guide-for-homeowners.html#:~:text=The%20cost%20of%20a%20simple,setup%20don't%20scale%20down.
2.2	Internal Works -Delivery and storage of all library items (£600 avg removal cost)	£600	£600	https://www.checkatrade.com/blog/cost-guides/house-removal-costs/
2.3	Services (Phase 3 works) -Removal of gas boiler, radiators (£2650 avg) -ASHP, UFH installation (ASHP avg cost £8000) (UFH £72p/m2) -LED installation -MVHR installation (£6900 avg)	£72p/m2 +£8000 +£6900 +£2650	£65,718	https://www.checkatrade.com/blog/cost-guides/cost-moving-boiler/ https://www.checkatrade.com/blog/cost-guides/air-source-heat-pump-cost/https://www.c heckatrade.com/blog/cost-guides/underfloor-heating-cost/https://www.myjobquote.co.uk/costs/installing-heat-recovery-system
2.4	External works -Ramp and concrete path (19.2m2, depth of 200mm)	£54p/m2	£1036.80	https://www.mybuilder.com/gardening-landscaping/price-guides/cost-of-pathway-installation
3	Building works estimate		£252,794.80	

An Elemental cost plan, subject to Main Contractor's preliminaries, OH&P VAT and inflation

Procurement Strategy

Design and Build Procurement

- The most common procurement method in the UK **accounting for over 50% of construction projects** (Designing Buildings. 2025)
- The single appointed Contractor will be allocated both design and construction risk. This provides a level of **Risk Safety** for yourselves as the client
- **Faster start time on site and in turn a faster project delivery** can be attained, as the Main Contractor can overlap early construction works with the design phase. For example, site prep and Phase 1 can begin, while design works for Phase 3-4 are still being finalised. (See Gantt Chart on [page x](#))



Alternative Strategy

- A traditional procurement strategy could be followed where an appointed design team refines the current designs to RIBA Stage 4, they would then be issued for tender and a contractor would be appointed.

Design and Build instead of Traditional

- Earlier start time on site
- Early contractor involvement provided greater cost certainty
- Risk allocation to a single contractor

Construction Programme: Gantt Chart

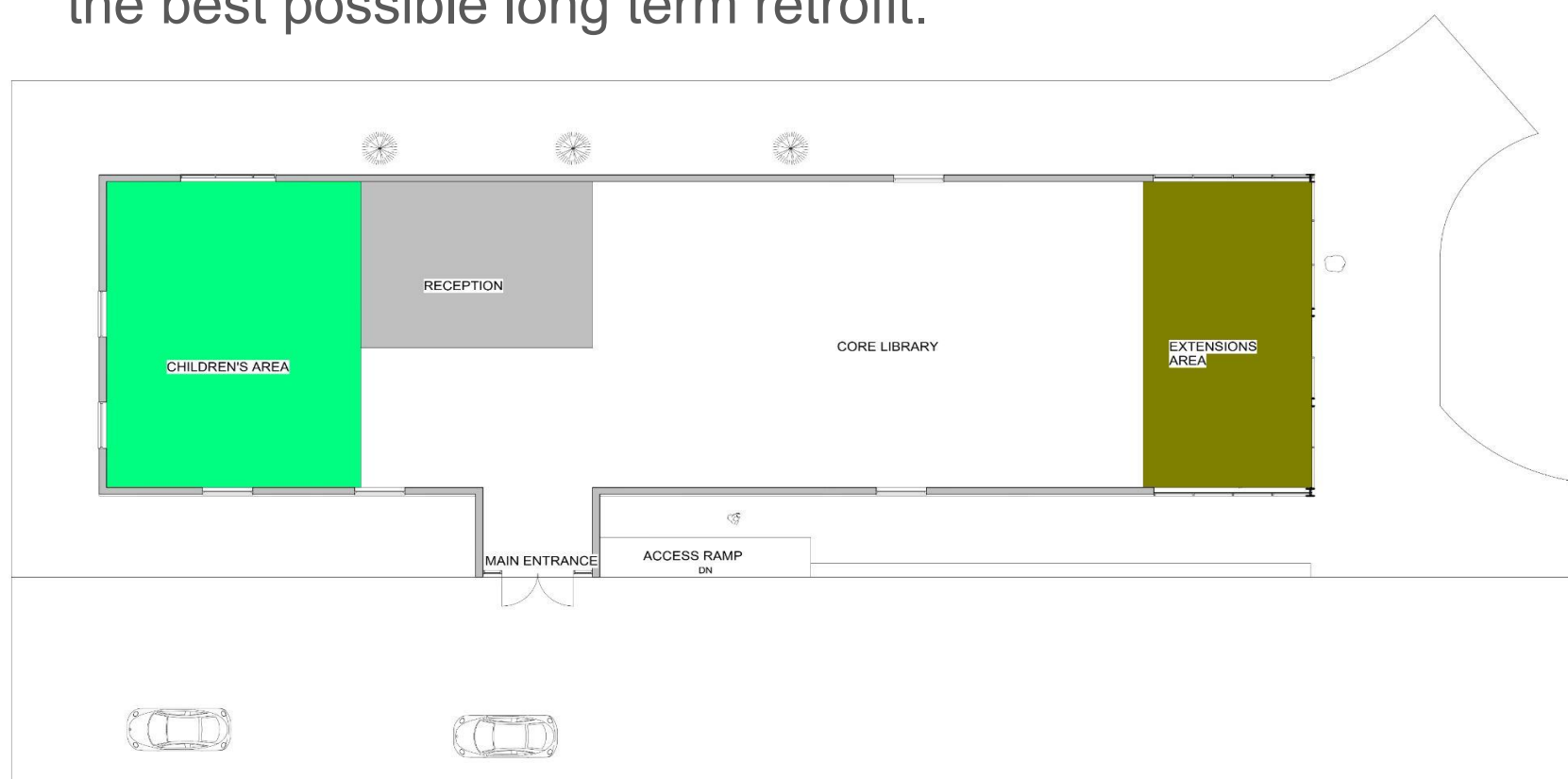
ACTIVITY	PLAN START	PLAN DURATION	PERIODS - Preliminary Start date 01/06/2026																																
			Total Planned Duration - 33 Weeks																																
			1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	17	18	19	20	21	22	23	24	25	26	27	28	29	30	31	32	33
Design stage	1	10																																	
Site Preparation	9	1																																	
Phase 1 - Extension	10	11																																	
Setting out	10	1																																	
Groundworks/ foundations prep	11	2																																	
Foundations	13	4																																	
Steel frame	17	2																																	
Subfloor and roof	19	1																																	
Glazing	20	1																																	
Phase 2 - External	19	4																																	
Redirection of path	19	2																																	
Ramp and Stairs	21	2																																	
Phase 3 - Internal	23	8																																	
Prep/Clearing library	23	1																																	
Eastern wall demolition	24	2																																	
Boiler/Radiators removal	26	0.5																																	
Subfloor removal UFH	26.5	1																																	
UFH/ASHP installation	27.5	2.5																																	
MVHR installation	30	1																																	
LED installations	30	1																																	
Phase 4 - Fabric	30	3																																	
Removal/Recycling old Single glazed panels	30	1																																	
Double glazing installation	31	1																																	
Compliance checks, Snagging, refurbishing	32	1																																	
Phase 5 - Handover/ Reopening	33	0																																	

26

Construction Programme: Phasing and Critical Paths

Programme phasing

- Construction programme phasing is key to delivering project in **32 weeks** and minimising library closure. Design and early construction works can overlap with a Design and Build Procurement strategy.
- During Phase 1 (11 weeks), the library can remain open while extension works are being carried out such as the substructure. Total closure of the library can therefore be **limited to 12 weeks.**
- The programme phases will overlap where interface is possible and minimise disruption/closure, while delivering the best possible long term retrofit.



Critical Path analysis

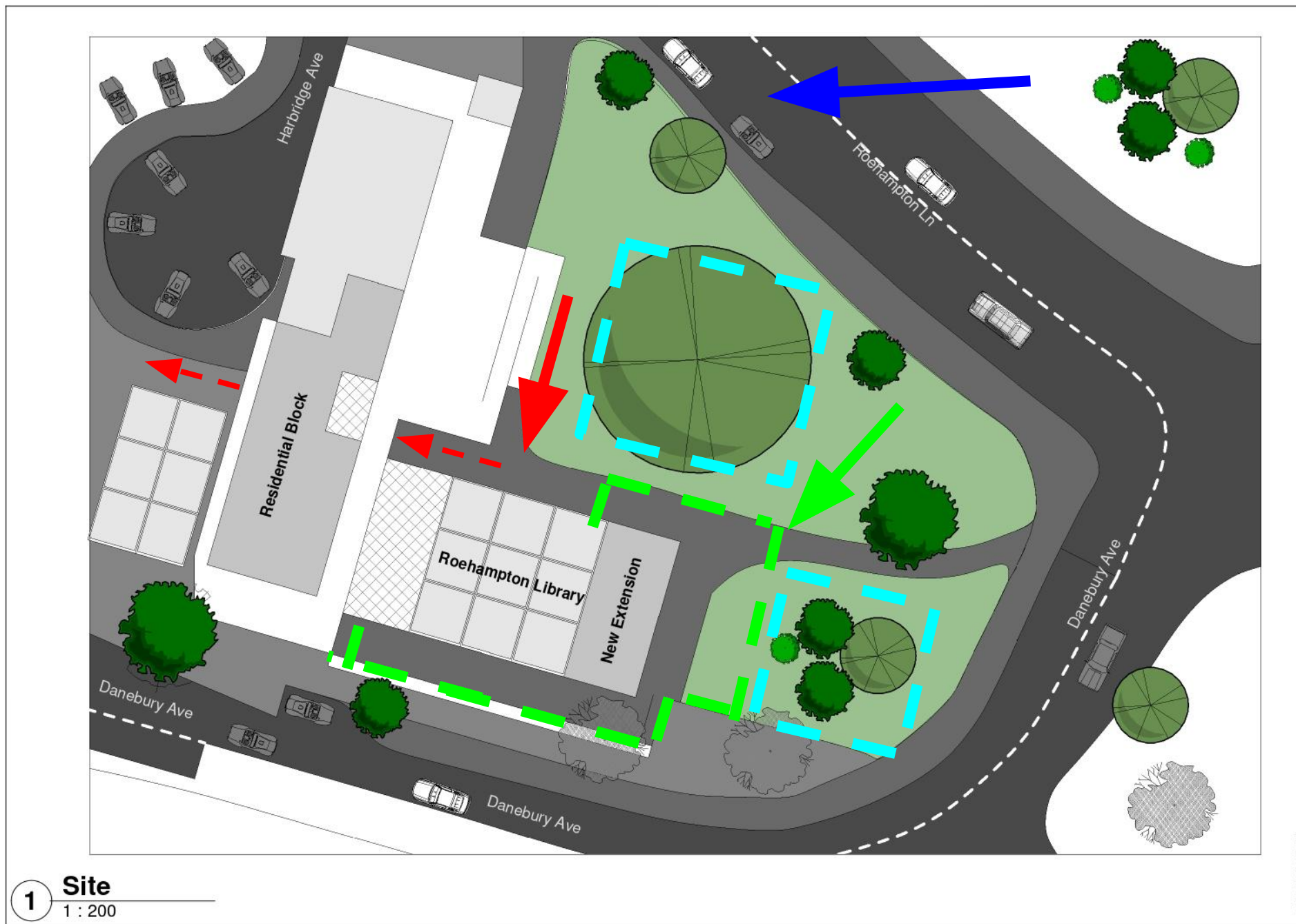
Certain works can overlap, such as:

1. The Phase 1 Extension and Phase 2 External works, as seen in the Gantt chart. Once the steel frame is assembled, creating the new residents path and ramp can be start.
2. The Phase 4 Double glazing can overlap with Internal works, as there will be minimal interface between external fabric and internal services.

Phase 1 Closure

- Due to noise from the Extension works, the Core library will close, however the children's area can remain open, as it is on the opposite side of the building

Construction Programme: Site Logistics Plan



Fencing and Site Entrance (Green)

- Site will be fenced off to create a safe distance from all extension and external works during **Phases 1-2**.
- The green arrow indicates the workers entrance
- This will allow adequate space for any material storage

Pedestrian Signage and Access (Red)

- Signage for pedestrians indicating the access path to Danebury Avenue from Roehampton Ln is closed and can be redirected via the western side of the library (underneath Albrook House)

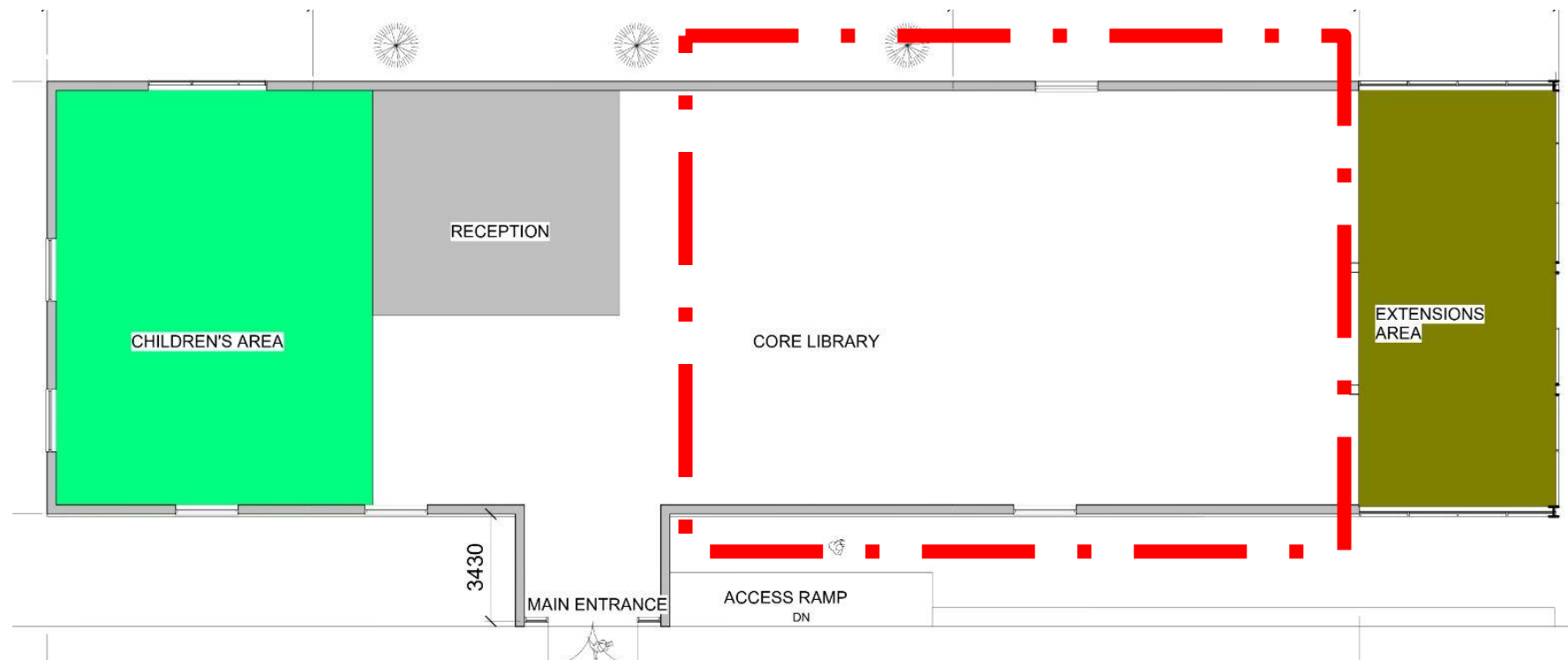
Tree Protection fencing (Blue)

- Fencing around any trees is required. All tree protection measures are subject to an Arboricultural survey

Deliveries and Storage (Blue)

- Permit holders parking can be requested for use by the council on Roehampton Lane for all large deliveries, the tight space and bus lane on Danebury avenue make this less possible.
- Large deliveries in **Phase 1-2** include, steel elements (Extension), and concrete deliveries for the foundations and external works
- During **Phase 3**, a pickup for all library items will be needed (then returned in **Phase 5**)

Considerations of Working in an Occupied Building



Phased Delivery

- The retrofit programme is tailored towards the local residents, community and all library users.
- The phased delivery over 32 weeks, ensures that total closure of the library is minimised.
- Due to noise from the extension works during Phase 1, the Core library will be closed, however the Children's area will remain open.

Considerations for Residents

- Mainly residents of Albrook House may be affected by noise and vibration from building works.
- All noise generating works during Phase 1-2 will be carried out during social hours to mitigate this.

Library Closure (Phase 1)

- Due to noise from the Extension works and for the safety of library users, the Core library (indicated by the **red square**) must be closed.

Library Closure (Phase 3-4)

- Complete closure needed for all service and fabric improvements. Library items will first be cleared. Closure of the library will be communicated to the community.

Health & Safety: Risk Assessment Matrix and Plan

Likelihood of Occurrence

Probable	4 Moderate	8 Major	12 Severe	16 Severe
Possible	3 Minor	6 Moderate	9 Major	12 Severe
Unlikely	2 Minor	4 Moderate	6 Moderate	8 Major
Rare	1 Minor	2 Minor	3 Minor	4 Moderate
	Low	Medium	High	Very High

Severity of Hazard

Risk Assessments

- Under Construction (Design and Management) CDM Regulations 2015, Risk assessments must be carried out for all construction activities
- Health and Safety is the responsibility of ourselves, the Contractor, as well as yourselves, the Client
- Information sourced from (HSE. 2015)

Risk Assessment Matrix and Plan

- For each construction phase (1-5) a RA can be carried out to identify a quantitative **Risk Rating** (Based on Likelihood and Severity)
- Hazards are identified and then given a Risk Rating based on the matrix
- **Control Measures** then act as risk mitigation strategies, put in place, and a **Risk Rating with Control Measures** can then be awarded

Health & Safety: Risk Assessment and Mitigation - Phase 1 (Extension)

Hazards	Risks, with no controls	Risk Rating	Control Measures to Reduce Risk	Risk Rating with Control Measures
Handling the Steel beams and Glazing	Strain injuries from lifting or getting hands caught between steel beams or glazing	8	Method statements in line with CDM (2015) regulations, careful lifting technique and use of lifting machinery where necessary	6
Falling objects such as steel bolts, beams or tools	Injuries that can vary in severity depending upon what object is, generally minor	6	Safety zones when a scaffold is erected preventing anyone being underneath	2
Structural instability, steel beams could collapse during construction	Without temporary works, structure could fall, severely endangering lives of those nearby	16	Design safety ensured by structural engineer, enabling works, as well as careful on site sequencing during frame assembly	4
Falls from height leading to injury	While installing the top steel beams, falls and slips could lead to injury, there will be a small scaffold (note that height is very small as it is a single story building)	6	Careful inspection of scaffold before use, workers are provided with method statements to carry out work on scaffold	2
Vibration and noise (To workers and to the public)	Installation of steel, use of tools will cause noise and vibration in a busy residential area	8	Ear protection provided to workers in immediate proximity (i.e. those using hand tools to install). Works kept to a short schedule during social hours	2
Slips, trips and falls	Workers are at risk of injury from falling if workspace is not clear or stable	4	Careful inspection of workspace and scaffold	2
Instability of scaffold	The scaffold could collapse if unstable, this would lead to a severe injury	8	Inspection of scaffold frame and foundation before use	6
Excavation for substructure of extension, including pouring pad concrete foundations	Risk of injury due to manual excavation, falling in excavated space and using tools. Concrete pouring also poses a risk of chemical burns or respiratory issues from inhaling silica dust	6	Excavation machinery can be used depending upon ground strength	2

Health & Safety: Risk Assessment and Mitigation - Phase 2-3 (External and Internal Works)

Hazards	Risks, with no controls	Risk Rating	Control Measures to Reduce Risk	Risk Rating with Control Measures
Manual lifting from clearing out library space, removing old windows from eastern wall removing gas boiler and radiators, concrete pouring and fitting heating and ventilation systems	Strain injuries due to manual lifting, significantly heavier items such as old windows and existing gas boiler	6	Method statements will be provided, workers should lift in pairs and use correct technique. Tools such as pallet trolleys can be used to transport items safely	3
Dust from demolition of eastern wall, pulling out subfloor (subject to asbestos checks)	Dust is very harmful to the lungs if inhaled excessively	6	Correct PPE is mandatory, respirators and eye protection for anyone inside the library	3
Gas leak, or in extreme case, carbon monoxide poisoning or explosion from removal of gas boiler	A gas leak or explosion, while unlikely, would have severe consequences to those working within the library as well as the nearby residents and public	16	Gas supply will be isolated prior to removal, removals will be carried out by certified gas safety engineers	4
Removing of old lighting to install new LEDs	Electrical hazards and use of stepladders could lead to electric shocks or slips and falls, resulting in injuries	4	Electrical supply will be isolated prior to removal, when using stepladders workspace and surface is carefully checked	2
Electrical hazards in installing ASHP and UFH	Electrical shocks and even serious injury	8	Electrical supply has been isolated	2
Concrete pouring, silica dust	Concrete works can cause chemical burns, skin irritation and respiratory issues	4	Mandatory PPE, gloves, long sleeves, respirators and eye protection	2

Health & Safety: Risk Assessment and Mitigation - Phase 4 (Building Fabric)

Hazards	Risks, with no controls	Risk Rating	Control Measures to Reduce Risk	Risk Rating with Control Measures
Manual handling of window units during removal and installation of new double glazed windows	Strain injuries from lifting and carrying window frames and panels	6	Careful phasing for each movements when lifting and installing frames, reducing twisting, prolonged carrying and awkward movements. Using trolleys/other equipment to help move panels	2
Sharp edges and dust	Injuries from cuts and inhalation of dust	6	Mandatory PPE, gloves, eye protection, dust masks/respirators when removing old frames	2

- While Control Measures will help to mitigate the risks, some Health & Safety risks will remain prevalent. In this instance it is important to be aware of them and be prepared for these activities.
- Method statements are issued to workers to provide clear instructions on carrying out the task, the safest way possible.

Health & Safety : CDM Regulations and Fire Safety

CDM Regulations 2015

The construction (Design and Management) Regulations 2015 establish the legal framework for construction safety in the UK.

- Risk Assessment Integration :

Proactive hazard identification at each project lifecycle stage.

- Duty Holder Responsibilities :

Designers, contractors, and clients share safety accountability.

- Principal Designer Role :

Coordinates health and safety during pre-construction phase.

- Principal Contractor Duties

Manages site safety during construction phase.

Fire and Life Safety

Retrofit project involving structural upgrades require robust fire safety measures compliant with Building Regulation Part B.

- Fire Risk Assessment

Comprehensive evaluation of fire hazards, ignition sources, and occupant risks

- Fire Detection Systems

Installation of smoke detectors, heat sensors, and alarm notification devices.

- Safe Escape Routes

Design of unobstructed egress paths with appropriate signage and lighting.

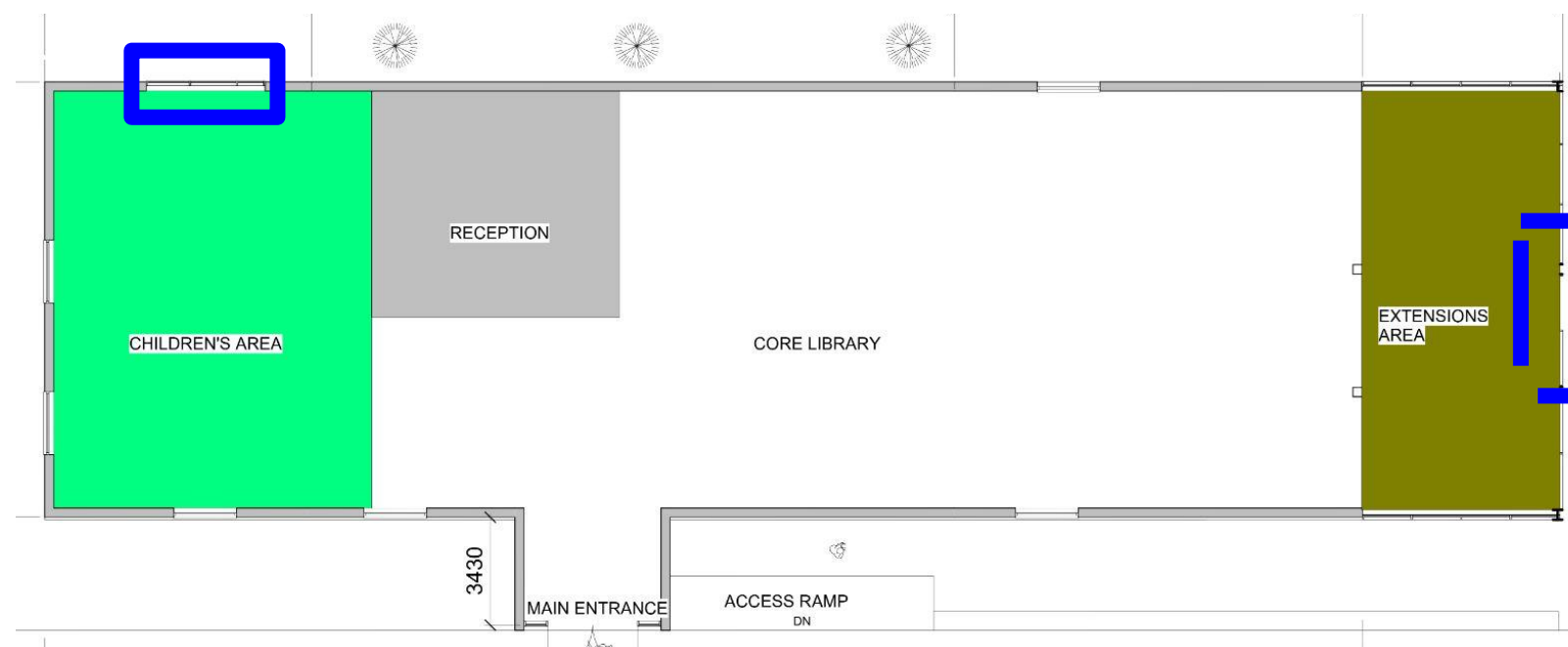
- ❖ Fire Resistant Materials

Specification of materials meeting fire resistance ratings (E.g, 30, 60, 90 minutes)

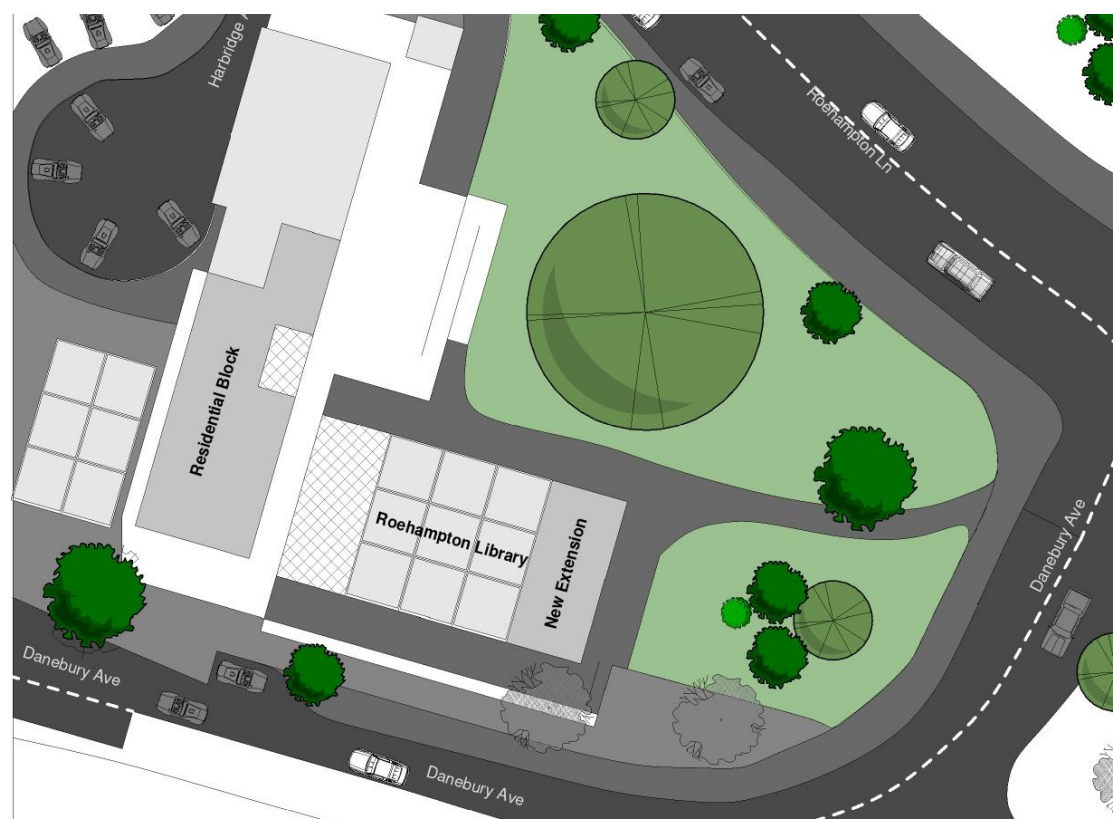
Fire & Life Safety Integration

Fire safety must be integrated early in the design process.

- Fire detection and suppression systems.
- Regular fire drills and evacuation procedures.
- Clear signage and illuminated escape routes.



Fire exits are indicated by blue squares



Fire Safety Plan during Construction

- Fire evacuation points must be signalled on the site during construction works
- Potential fire risks can occur during the internal works, removing electrical equipment and gas boiler

Construction Methodology

Phased Approach Framework

- A systematic phased approach ensures safety, efficiency, and minimal disruption during retrofit projects. This methodology divides complex work into manageable stages.

Site Preparation Phase



- Comprehensive hazard assessments, establishment of safety barriers, material staging areas, and site access control protocols. Critical for establishing safe working conditions.

Execution of Retrofit Works



- Implementation of structural upgrades, thermal insulation installation, and building systems improvement. Requires coordination across multiple trades and disciplines.

Post Retrofit Checks



- Verification that all work meets safety and regulatory standards before handover. Includes commissioning, testing, and documentation for compliance.



Complexity in Retrofitting

Retrofitting involves working with existing structures that may not meet current codes, requiring.

- Adaptive methodologies for unknown conditions
- Structural assessment and reinforcement.
- Integration of new systems with old.
- Compliance with evolving regulations.

Construction Methodology Phased Approach

Pro-Construction

Site Surveys

- Comprehensive assessment of existing conditions, structural integrity, and potential hazards

Permit Acquisition

- Obtaining planning permission, building control/approval, and CDM compliance documentation

Safety Barriers

- Establishing physical boundaries, signage, and access control system.

Worker Induction

- Site-specification safety training, hazard awareness, and emergency procedures.

Post Construction

Safety Audits

- Thorough inspection of all completed work against safety standards and CDM requirements

Compliance Checks

- Verification of Building Regulations Part B (fire safety), Part L and Part P (electrical)

System Integration

- Final commissioning of MEP systems, testing, and handover documentation

Operational Safety

- Ensuring all systems are operational, safe, and meet client requirements

Temporary Work Considerations Overview

Worker Access



- Safe platforms and pathways for personnel at height or in confined spaces

Structural Stability



- Temporary support preventing collapse during structural modifications

Material Handling



- Support for construction materials and equipment loads

Critical Importance

- Working at Height Protection: Preventing falls from elevated work positions during facade or roof work
- Unstable Structure Management : Supporting weakened elements during load transfer or removal.
- Hazardous Material Containment : Isolating asbestos removal zones and chemical work areas

Hazard Containment



- Barriers preventing unauthorised access to dangerous work zones



Temporary Works Scaffolding, Shoring & Safety Barriers

Scaffolding systems

Provide safe access to building areas undergoing retrofit, particularly critical for high-rise projects.

- Design Requirements: Engineered to support worker weight, materials, and equipment loads with safety factors
- Erection Standards: Competent person only, compliance with TG2013 and NASC guidance.
- Inspection Protocol : Weekly inspections by qualified scaffolders, after adverse weather or incidents.
- Types: Independent, putlog, system, and suspended scaffolds for different applications.

Shoring systems

- Provide temporary structural support during retrofit operations
- Types: Raking shores, flying shores, dead shores, and hydraulic systems.
- Monitoring: Regular inspection for settlement, movement, or distr

Safety Barriers

- Safety barriers establish clearly defined boundaries around hazardous work areas.
- Signage and fencing around exclusion zones such as excavated areas and scaffolding

Risk Identification

Retrofit project require systematic identification across four critical risk categories, Each category demands specific assessment methodologies and mitigation strategies

Health & Safety

- Physical injury from machinery
- Electrical hazards from live circuits
- Toxic material exposure (asbestos)
- Falls from heights
- Manual handling injuries

Financial

- Project delays and extensions.
- Unforeseen expenses
- Material cost fluctuations
- Scope creep
- Contractor insolvency

Environmental

- Energy consumption during works
- Wast generation and disposal
- Emissions and air quality
- Noise Pollution
- Water contamination

Structural

- Existing structure integrity.
- Load bearing capacity
- Foundation stability
- Material degradation
- Compatibility with new systems

Ground Conditions

Site: Roehampton Library, Danebury Avenue, SW15 4AR

Data Source: British Geological Survey (BGS)

Top Layer – Head Deposits

- Clay, silt, sand and gravel
- Variable material
- Not suitable for foundations

Below – London Clay Formation

- Stiff clay and silt
- Strong when dry
- Expands and shrinks with moisture

What This Means for Foundations

- Foundations should be placed in London Clay
- Recommended depth: **1.5 m**
- Consider shrink–swell effects

Foundation Proposal

Project: Single-Storey Library retrofit

Site: Roehampton Library, SW15 4AR

Ground Conditions: Head Deposits over London Clay (BGS)

Proposed Foundation: Reinforced concrete **pad foundation** are recommended for the proposed single story extension of Library

Justification

- Head deposits are variable and unsuitable for bearing.
- London Clay provides a stiff, competent bearing layer.
- Steel columns carry concentrated loads, making pad foundations efficient and economical.

Design Consideration

- Found in London Clay (avoid head deposits)
- Consider shrink–swell effects
- Assess bearing capacity and settlement

Pad foundations should be:

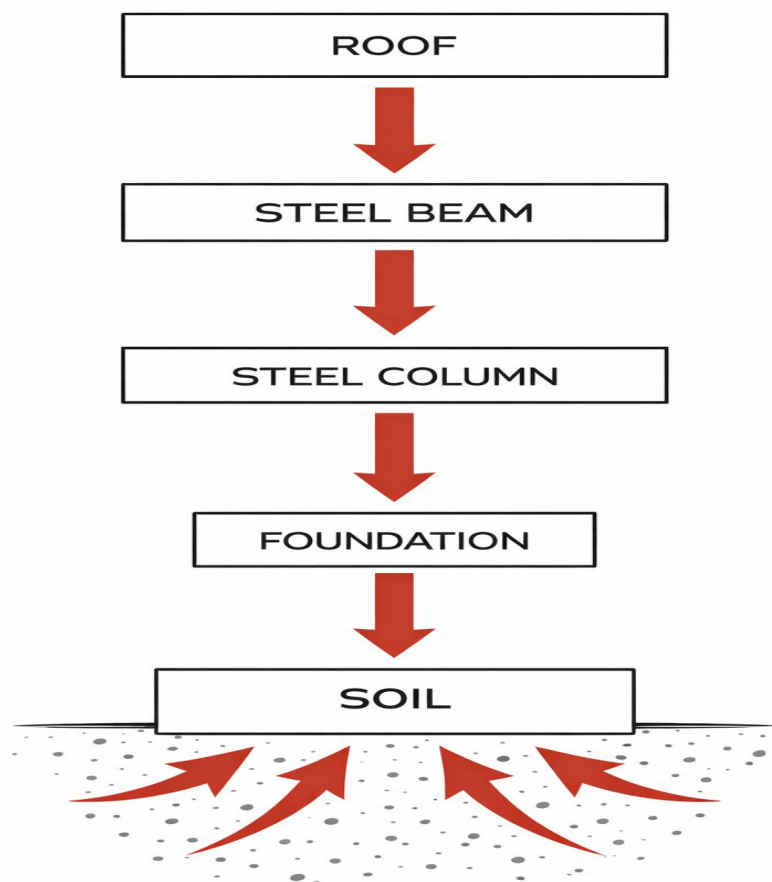
- Founded within London Clay at approximately 1.5 m below ground level

This ensures:

- Foundations are below the variable head deposits
- Reduced impact from seasonal moisture changes
- Improved bearing performance

Load path and structural Retrofit Impact

This Diagram shows load transfer path for the proposed Roehampton retrofit.



- ❑ The roof slab carries dead and imposed load such as self and snow weight.
- ❑ These load are transferred vertically to the steel beam.
- ❑ The beam then transfers the load to steel column.
- ❑ The column carries the load to the reinforced foundation.
- ❑ Finally, the foundation spread the load into the soil distributing the pressure safely into london clay.

Structural Retrofit Impact

- ❑ Boosts the load-bearing ability of current structural components
- ❑ Enhances the safety and stability of the structure when loads are applied
- ❑ Increases the durability and lifespan of the building
- ❑ Lowers deflection, cracking, and deterioration of the structure
- ❑ Strengthens resistance to extreme events

Load and slab Calculation

1

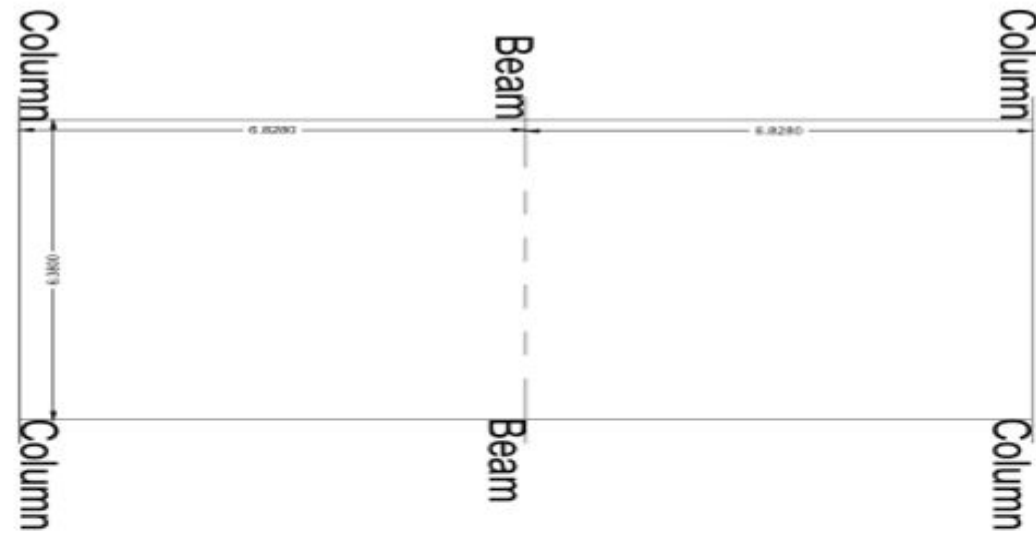


Fig: Layout of beam and column

References	Calculation	Outcomes
Cobb's Fiona book pg. 93 C11	Load Calculation	
Bauder.co.uk	Live Load (Q_k) = 2 Kn/m^2 Load of water proofing Mass per unit = $5 \text{ kg/m}^3 = 0.05 \text{ Kn/m}^2$ Thickness = 4mm Weight of water proofing = $0.05 \times 4 = 0.2 \text{ kn/m}^2$	
1991-1-3 cl 5.1 1991-1-3 T 5.1 1991-1-3 T 5.2 1991-1-3 pg.20 Dulbal software	Calculation of a snow load Snow load = $\mu_i \cdot C_e \cdot C_t \cdot S_k$ $\mu_i = 0.8$ $C_e = 1$ $C_t = 1$ $S_k = 0.35 \text{ kn/m}^2$ (snow zone 3) Snow load = $0.8 \times 1 \times 1 \times 0.35 = 0.28 \text{ kn/m}^2$	
1991-1-4 Note 2 1991-1-4 Note 3	Calculation of Wind Load Basic wind velocity (V_b) = $C_{dir} \cdot C_{seasonal} \cdot V_{bo}$ $C_{dir} = 1$ $C_{seasonal} = 1$ $V_{bo} = 24 \text{ m/s}$ (Dlubal Software) Now, $V_b = 1 \times 1 \times 24.5 = 24.5 \text{ m/s}$	

References	Calculation	Outcomes
1991-1-4 Cl 4.5 1991-1-4 Cl 4.4 1991-1-4 Cl 4.3	Mean wind velocity (V_m) = $C_r \cdot C_o \cdot V_b$ $C_r = 0.8$ $C_o = 1$ Now, $V_m = 0.8 \times 1 \times 24.5 = 19.6 \text{ m/s}$ Velocity Pressure (q_p) = $0.5 \rho v_m^2$ $= 0.5 \times 1.25 \times 19.6^2$ ($\rho = 1.25 \text{ kg/m}^3$) $= 240 \text{ N/m}^2 = 0.24 \text{ N/mm}^2$	
1991-1-4 T 7.2	Pressure coefficient for flat roof $C_p = -0.9$ [Negative because wind cause uplift] Roof area of extension = $13.68 \times 6.38 = 63.6 \text{ m}^2$ Wind load on a roof = $0.24 \times 0.9 = 0.21 \text{ kn/m}^2$	
	Load from the doors = $1.98 \times 0.9 \times 0.04 \times 4.1$ $= \frac{0.29}{1.7} = 0.16 \text{ kn/m}^2$ Load from the glazing = $25 \times 0.008 = 0.2 \text{ kn/m}^2$ Load from suspended ceiling = 0.50 kn/m^2 Total dead Load (G_k) = $5.75 + 0.16 + 0.2 + 1 + 0.50 = 7.1 \text{ kn/m}^2$ Total live load (Q_k) = $2 + 0.28 + 0.21 = 2.85 \text{ kn/m}^2$	DL = 7.1 kn/m^2 L.L = 2.85 kn/m^2
	Slab Calculation Longer length (l_y) = 13.646 m Shorter side (l_x) = 6.38 m Now, Checking for type of slab $\frac{l_y}{l_x} = \frac{13.646}{6.38} = 2.13$ Hence, it's greater than 2, so it's a one-way slab. Again, Overall depth of a slab = $L/24 = 6380/28 = 230 \text{ mm}$ Floor finish = 1 kn/m^2 Self-weight of a slab = $230 \times 25 \times 10^{-3} = 5.75 \text{ kn/m}^2$ Ultimate load = $1.5 Q_k + 1.35 G_k$ $= 1.5 \times 2.85 + 1.35 \times 7.1 = 13 \text{ Kn/m}^2$	d = 194 mm D = 230 mm
BS EN 1992-1 Table 4.5N	Minimum cover = 30 mm (considering xc4 exposure class S2 structural class allowing say 6 mm as a half diameter of a reinforcement. Effective Depth (d) = $230 - 6 - 30 = 194 \text{ mm}$	Cover = 30 mm H12 Reinforcement

2

Continuation of slab Calculation

3

References	Calculation	Outcomes
BS EN 1992-1 Table:3.1	Bending Moment(M_{ed}) = $wl^2/8$ $= 13 \times 6.38^2 / 8 = 67KNm$ Bending Reinforcement (k) = $\frac{M}{bd^2 f_{ck}}$ $= \frac{67 \times 10^6}{1000 \times 194^2 \times 25} < 0.167 = 0.07 (ok)$ We assume c25 concrete class Determining lever arm, $z = d \left[0.5 + \sqrt{0.25 - \frac{k}{1.134}} \right]$ $= d \left(0.5 + \sqrt{0.25 - \frac{0.007}{1.134}} \right)$ $= 0.99d \text{ (which is greater than } 0.95d \text{) so,}$ $A_s = \frac{M}{0.87 \cdot f_{yk} \cdot 0.95d} = \frac{67 \times 10^6}{0.87 \times 500 \times 0.95 \times 194} = 832mm^2$ Provide H12 bars at 125mm spacing where $A_s = 905mm^2$ Checking span effective depth ratio $\rho = \frac{100 \cdot A_s}{bd} = \frac{100 \times 905}{1000 \times 194} = 0.42\% > 0.15\% \text{ min requirement}$ Minimum and maximum Reinforcement $A_{s \min} = 0.26 \left(\frac{f_{ctm}}{f_{yk}} \right) bd$ $= 0.26 \left(\frac{2.6}{500} \right) 1000 \times 194 = 263mm^2$ Provide H12 bars at 300mm spacing $A_s = 377mm^2$ For SHEAR Design shear force $V_{ed} = \frac{wl}{8} = \frac{13 \times 6.38}{8} = 11kn/m$ $V_{Rdc} = [0.12k (100\rho_1 f_{ck})^{1/3}] bd$ Now, $K = (1 + 200/d)^2 = (1 + 200/194)^{0.5} = 2.01 \leq 2 (ok)$	$A_s = 905mm^2$ H12 bars 125 mm of space $A_s = 377mm^2$ H12 bars, 300mm spacing

References	Calculations	Outcomes
BS EN 1992-1:2004 EN 1992-1-1:2004 Table:7.4N	$\rho_1 = \frac{A_s}{bd} = \frac{905}{1000 \times 194} \leq 0.02 = 0.004 (ok)$ $V_{Rdc} = [0.12k (100\rho_1 f_{ck})^{1/3}] bd$ $= [0.12 \times 2 (100 \times 0.004 \times 25)^{1/3}] 1000 \times 194 = 100.3KN$ $V_{min} = [0.035K^{1.5} \cdot f_{ck}^{0.5}] \cdot bd$ $= [0.035 \times 2^{1.5} \cdot 25^{0.5}] 1000 \times 194 = 96.04kN$ $V_{Rdc} = 100.31KN > V_{ed} = 11kn (ok)$ Hence, it passes the shear check. Deflection check $\rho = \frac{A_{s,req}}{bd} = \frac{832}{1000 \times 194} = 0.004$ Reference reinforcement ratio, $\rho_0 = f_{ck} \times 10^{-3} = 25 \times 10^{-3} = 0.025$ For structural system $k=1$ $(l/d)_{basic} = K \left[11 + 1.5\sqrt{f_{ck}} \frac{\rho_0}{\rho} + 3.2\sqrt{f_{ck}} \left(\frac{\rho_0}{\rho} \right)^{1.5} \right]$ $= 1[11 + 22 + 2] = 35$ Modification factor = 1.0 Modification factor for steel area provided = $A_{s,prov}/A_{s,req}$ $= 905/832 = 1.08 < 1.5 (ok)$ Therefore, allowable span-effective depth ratio: $(L/D)_{allowable} = 35 \times 1.08 \times 1 = 38$ Actual span effective depth-ratio: $(L/D)_{actual} = \frac{6380}{194} = 32.9 < (L/D)_{allowable} (ok)$	$V_{Rdc} = 100.31KN$

Continuation of slab & beam Calculation

References	Calculation	Outcomes
	Calculation for a steel Beam Total live load (Q_k) = 2.85 kn/m² Total Dead Load (G_k) = 6.75 kn/m² Now, $f_{ed} = 1.5Q_k + 1.35G_k$ $= 1.5 \times 2.85 \times \frac{6.38}{2} + 1.35 \times 6.75 \times \frac{6.38}{2}$ $= 42 \text{ kn/m}$ Design Bending moment (M_{ed}) = $\frac{42 \times 6.38^2}{8} = 213 \text{ knm}$ Design Shear Force = $\frac{wl}{2} = \frac{42 \times 6.38}{2} = 133 \text{ knm}$ Cross-section selection: assume suitable section belongs to class 1 We considered the S 500 steel ($f_y = 500 \text{ N/mm}^2$) Minimum required plastic moment of resistance about the major axis (y-y), Wply, is given by: $Wply \geq \frac{M_{pl} R_d}{f_y} = \frac{M_{ed}}{f_y} = \frac{213 \times 10^6}{500} = 426 \times 10^3 \text{ mm}^3$	Wply=426 × 10 ³ mm ³

5

7

References	Calculation	Outcomes
	Actual Deflection $\delta_{max} = \frac{5wl^4}{384EI}$ $= \frac{5 \times 31.29 \times 6380^4}{384 \times 210000 \times 141.4 \times 10^6} = 22 \text{ mm}$ Serviceability check $\frac{\delta_{max}}{\delta_{allowable}} = \frac{22}{25} = 0.8 < 1 \text{ (ok)}$ Hence, 356 × 171 × 51 UB Provided is adequate.	

References	Calculations	Outcomes
Steel design data pg. B4 BS EN 1993-1-1:2005 BS 4-1:2005 Table 14.1 BS EN 1993-1-1	Depth of a section = 355mm Width of a section = 171mm Web thickness = 7.4mm Flange thickness = 11.5 mm Root Radius = 10.2 mm Area = 64.9 Cm² Try 356 × 171 × 51 Wply = 896Cm³ Checking the classification of cross-section $\epsilon = \sqrt{\frac{235}{f_y}} = \sqrt{\frac{235}{500}} = 0.68$ $\frac{C}{t_f} = \frac{(b - t_w - 2r)/2}{t_f} = \frac{(171 - 7.4 - 2 \times 10.2)/2}{11.5} = 6.2 \leq 10\epsilon = 6.8 \text{ (class 2)}$ $\frac{d}{t_w} = \frac{355}{11.5} = 47 \leq 72 \epsilon$ $= 47 \leq 72 \times 0.68$ $= 47 \leq 48.96 \text{ (Class 1)}$ 356 × 171 × 51 UB Belongs to class 2 as assumed Resistance of cross section to Bending Moment $M_{pl} R_d = Wply \times f_y$ $= 896 \times 10^3 \times 500 = 448 \text{ knm}$ $\frac{M_{ed}}{M_{pl} R_d} = \frac{213}{448} \leq 1 = 0.4 \text{ (ok)}$ Resistance of cross-section to shear $V_{PLRD} = \frac{A_v \times f_y}{\sqrt{3}} = \frac{[A - 2bt_f + (t_w + 2r)t_f] \times f_y}{\sqrt{3}}$ $= \frac{[6490 - 2 \times 171 \times 11.5 + (7.4 + 2 \times 10.2) \times 11.5] \times 500}{\sqrt{3}} = 830$ Now, $\frac{V_{ed}}{V_{PLRD}} = \frac{133}{830} = 0.16 \leq 1 \text{ (ok)}$ Vertical deflection at serviceability Allowable Deflection $\delta_{allowable} = \frac{\text{span}}{250} = \frac{6380}{250} = 25 \text{ mm}$	UB class2 sections $V_{PLRD} = 830$

6

Foundation Design

References	Calculation	Outcomes
Cobb's Fiona book pg.329	Foundation Design We assumed a concrete class C30 and S235 steel in accordance with Eurocode Ultimate load = 13kn/m^2 Area= 87.02 m^2 Now, Total load (N_{ed}) = $13 \times 87.02 = 1131\text{kn}$ We have 6 column so, Bearing capacity of a soil= 100kpa Load per column = $\frac{1131}{6} = 188\text{ kn}$ Now, Required footing Area= $\frac{\text{load}}{\text{bearing capacity of soil}} = \frac{188}{100} = 1.87\text{kn}$	Bearing capacity = 100kn
	Assume Square footing, Side= $\sqrt{1.87} = 1.36 \approx 1.5\text{m}$, Assume 1.5×1.5 Soil pressure = $\frac{188}{1.5^2} = 83.5 < 100$ (ok) Earth pressure= $\frac{1131}{1.5^2} = 502\text{kn/m}^2$ Assume a 500 mm footing clear cover is 40 mm based on xc2 exposure class and S2 structural class. So effective depth is 462mm and half of dia is 8mm At the column face, $= ud \left[0.6 \left(1 - \frac{f_{ck}}{250} \right) \right] \frac{f_{ck}}{1.5}$ $= 0.5 \times 6 \times 152 \times 462 \left[0.6 \left(1 - \frac{30}{250} \right) \right] \frac{30}{1.5} \times 10^{-3}$ $= 2215 > N_{ed}$	Square Footing size=1.5m

References	Calculation	Outcomes
Mosley et al,2012 Table:8.2	Punching shear Basic control perimeter= $\text{column perimeter} + 4\pi d$ $= 6 \times 152 + 4 \times 3.14 \times 460$ $= 6692\text{mm}$ Area within perimeter= $(152 + 4d)^2 - (4 - \pi) \cdot (2d)^2$ $= (152 + 4 \times 462)^2 - (4 - \pi) \cdot (2 \times 460)^2$ $= 3.5 \times 10^6\text{mm}$ Punching shear force (V_{ed}) = $502(3.5 - 1.5^2) = 627\text{kn}$ Punching shear stress= $\frac{V_{ed}}{\text{perimeter} \times d}$ $= \frac{627 \times 10^3}{6692 \times 460} = 0.2\text{ N/mm}^2$ The ultimate shear stress is not excessive therefore h=500mm be a suitable estimate Bending Reinforcement At the column face, which is critical section $M_{ed} = \left(502 \times 1.5 \times \frac{1.5 - 0.152}{2} \right) \times \frac{(1.5 - 0.152)/2}{2}$ $= 501.5\text{kn}$	$M_{ed} = 501.2\text{kn}$
	for the concrete, $M_{bal} = 0.167 f_{ck} b d^2$ $= 0.167 \times 30 \times 1500 \times 462^2 = 1325\text{knm} > 501.5$ $A_s = \frac{M_{ed}}{0.87 \times f_{yk}}$ from the lever arm curve figure 4.5 $l_a = 0.95$ $A_s = \frac{501 \times 10^6}{0.87 \times 235 \times 0.95 \times 462} = 1382\text{ mm}^2$ Provide H16 bars at 100 mm spacing $A_s = 2010\text{mm}^2$ $\text{steel area}_{req} = 0.26 \frac{f_{ctm}}{f_{yk}} = 0.26 \times \frac{2.9}{235} = 0.27$ $\text{Steel area}_{prov} = \frac{100 A_s}{bd} = \frac{100 \times 2010}{1500 \times 460} = 0.29$ Hence, the steel provided is adequate Maximum bar size Steel stress should be calculated under the action of the quasi-permanent loading which can be estimated from equation 6.1 as follow= $\frac{f_{yk}(G_k + 0.3Q_k)}{1.15(1.35G_k + 1.5Q_k)}$ $= \frac{235(430 + 0.3 \times 158)}{1.15(1.35 \times 430 + 1.5 \times 152)} = 63\text{N/mm}^2$ Maximum bar size is 16 mm; crack control requirements are satisfied	100 mm spacing H16 bars $A_s = 2010\text{mm}^2$

References	calculation	Outcomes
Mosley et al,2012 T8.2	Final check of punching shear The shear resistance of the concrete without shear reinforcement can be obtained from table 8.2 where $\rho_1 = \frac{A_s}{bd} = \frac{2010}{1500 \times 462} = 0.29\%$ Here, $V_{Rdc} = 0.58\text{ N/mm}^2$ Here shear resistance of concrete $(V_{Rdc}) = V_{Rdc} \times \text{basic control perimeter} \times d$ $= 0.58 \times 6692 \times 462 = 1208\text{kn} > V_{ed} (627\text{kn})$ Maximum shear force At critical section for shear 1.0d from column face Design shear force (V_{ed}) = $502 \times 1.5 \times (0.67 - 0.152) = 390\text{kn}$ Now, $V_{Rdc} = V_{Rdc} b d = 0.58 \times 1500 \times 460 \times 10^{-3}$ $= 400\text{kn} > V_{ed} (275\text{kn})$ No shear reinforcement is required. All relevant design checks have been satisfied; hence, the foundation design is adequate.	$V_{ed} = 390\text{kn}$ $V_{Rdc} = 400\text{kn}$



Fig:Firm clay under finger pressure

As stated in Cobb's Civil Engineering Handbook (p. 329), if the soil can be shaped easily with firm finger pressure, it is categorized as firm clay (London clay), and a permissible bearing capacity of about 100 kPa can be used. This reflects a conventional field evaluation technique, where the soil's bearing capacity is judged through visual inspection and basic manual tests, without the need for formal lab or in-situ testing.

Column Design

References	Calculation	outcome
Steel Building Design Data	Design of a steel column $N_{ed} = 188kn$ $f_y = 235mpa$ $\gamma_{m1} = 1$ $Length = 3m$	
	End condition pin $l_{eff} = 3m$ Select a trial column section try UB 152 × 152 × 23	
	Geometric properties $Area\ of\ a\ section = 29.2cm^2$ $i_y = 6.54cm$ $i_z = 3.50cm$ $thickness\ of\ flange = 6.8mm$ $\frac{c_f}{t_f} = 9.65$ $\frac{c_w}{t_w} = 21.3$	
BS EN 1993-1-1 Table 5.2	Check cross section classification under the pure compression Need only check that section is not class 4 For outstand flange $c_f/t_f \leq 14$ For web $c_w/t_w \leq 42$	$N_{c,Rd}$ = 686kn
BS EN 1993-1-1 Cl 6.2.4	$\epsilon = \sqrt{\frac{235}{f_y}} = \sqrt{\frac{235}{235}} = 1.0$ Actual $c_f/t_f \epsilon = 9.65 \times 1 = 9.65$; within limit Actual $c_w/t_w \epsilon = 21.3 \times 1 = 21.3$; within limit ∴ Section is not Class 4	
	Cross-section compression resistance: $N_{c,Rd} = \frac{A f_y}{\gamma_{M0}}$ $= \frac{2920 \times 235}{1.0} \times 10^{-3} = 686\ kn > 188\ kn = N_{ed}$ OK Effective lengths: $L_{cr,y} = 1.0L = 1.0 \times 3000$ $= 3000mm$ (buckling for y axis) $L_{cr,z} = 1.0L = 1.0 \times 6380$ $= 6380\ mm$ (buckling about the z-z axis)	

10

References	Calculation	outcomes
BS EN 1993-1-1 Table :6.2	Non-dimensional slenderness $\lambda_1 = \pi \sqrt{\frac{E}{f_y}} = \pi \sqrt{\frac{210000}{235}} = 93.91$ $\lambda_y = \frac{\lambda_y}{\lambda_1} = \frac{L_{cr,y}/i_y}{\lambda_1} = \frac{3000/65.4}{93.91} = 0.48$ $\lambda_z = \frac{\lambda_z}{\lambda_1} = \frac{L_{cr,z}/i_z}{\lambda_1} = \frac{6380/35}{93.91} = 1.94$	$\lambda_y = 0.48$ $\lambda_z = 1.94$
	Buckling curves: For major axis buckling, use buckling curve 'b' For minor axis buckling, use buckling curve 'c'	
	BS EN 1993-1-1 Cl.6.3.1.2 $\phi_y = 0.5 [1 + \alpha(\lambda_y - 0.20 + \lambda_y^2)]$ $= 0.5[1 + 0.34(0.48 - 0.2) + 0.48^2] = 0.9$ $\phi_z = 0.5 [1 + \alpha(\lambda_z - 0.20 + \lambda_z^2)]$ $= 0.5[1 + 0.49(1.94 - 0.2) + 1.94^2] = 2.3$	$\phi_y = 0.9$ $\phi_z = 2.3$
BS EN 1993-1-1 Cl.6.3.1.2	$x_y = \frac{1}{\phi_y + \sqrt{\phi_y^2 - \lambda_y^2}}$ $= \frac{1}{0.9 + \sqrt{0.9^2 - 0.48^2}} = 0.6 \leq 1$	$x_y = 0.6$ $x_z = 0.28$
	$x_z = \frac{1}{\phi_z + \sqrt{\phi_z^2 - \lambda_z^2}}$ $= \frac{1}{2.3 + \sqrt{2.3^2 - 1.94^2}} = 0.28 \leq 1$	
	Buckling resistances: $N_{b,y,Rd} = \frac{\chi_y A f_y}{\gamma_{M1}} = \frac{0.6 \times 2920 \times 235}{1.0} \times 10^{-3}$ $= 412\ kn > 188\ kn = N_{ed}$ (OK) $N_{b,z,Rd} = \frac{\chi_z A f_y}{\gamma_{M1}} = \frac{0.28 \times 2920 \times 235}{1.0} \times 10^{-3}$ $= 192\ kn > 188\ kn = N_{ed}$ (OK) Hence 152 × 152 × 23 UC is adequate	$N_{b,y,Rd}$ = 412kn $N_{b,z,Rd}$ = 192kn

11

Fire and structural robustness for beam, column and slab

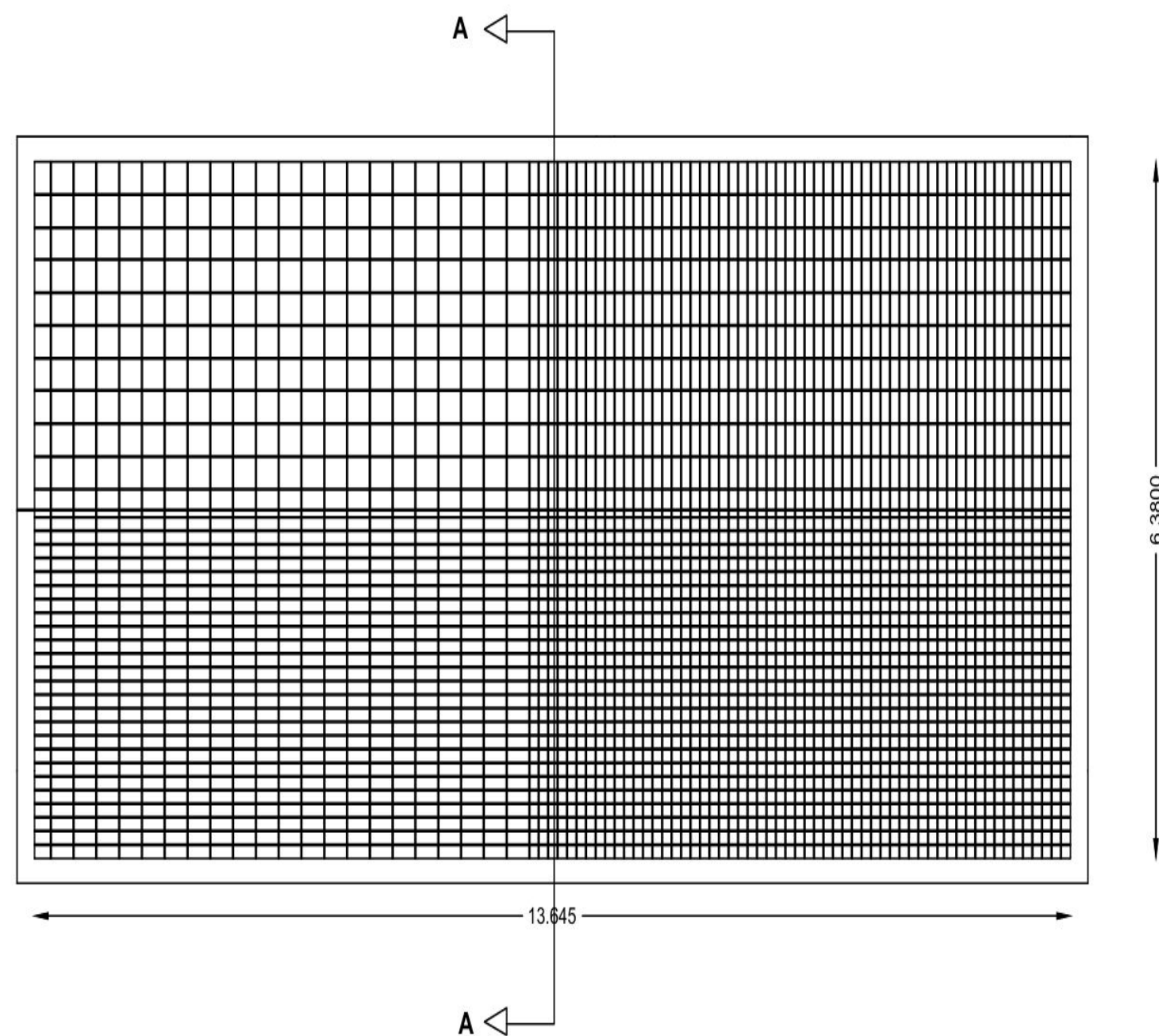
12

References	Calculation	Outcomes
BS EN 1993-1-2:2005 Cl. 2.4.2	Fire and structural robustness considerations Steel beam fire check From your beam design: $M_{Ed} = 213 \text{ kNm}$ $M_{pl,Rd} = 448 \text{ kNm}$	
BS EN 1993-1-2:2005 Cl. 4.2.4	Load ratio: $\mu_0 = \frac{M_{Ed}}{M_{pl,Rd}} = \frac{213}{448} = 0.48$ Using the simplified critical temperature expression from EN 1993-1-2: $\theta_{a,cr} = 39.19 \ln \left(\frac{1}{0.9674 \mu_0^{0.333}} + 1 \right) + 482$ $\theta_{a,cr} = 39.19 \ln \left(\frac{1}{0.9674 (0.48)^{0.333}} + 1 \right) + 482$ $\theta_{a,cr} \approx 596^\circ \text{C}$ For 60-minute fire resistance, the beam satisfies provided their steel temperatures do not exceed 596 C	
BS EN 1991-1-7:2006 Cl. 3.3	Robustness of steel beam Robustness is checked using the tying rules in BS EN 1991-1-7. A minimum tie force of 75 kN is commonly used for horizontal ties in steel-framed buildings. Beam robustness check Beam gross area from your section: $A = 64.9 \text{ cm}^2 = 6490 \text{ mm}^2$	
BS EN 1993-1-1:2005 Cl. 6.2.4	Using steel grade S500: $N_{pl,Rd} = \frac{A f_y}{\gamma_{M0}}$ Taking $\gamma_{M0} = 1.0$: $N_{pl,Rd} = \frac{6490 \times 500}{1.0} \times 10^{-3}$ $N_{pl,Rd} = 3245 \text{ kN}$	
BS EN 1993-1-2:2005 Cl. 2.1(1)	Check against tie force: $3245 \text{ kN} > 75 \text{ kN}$ The beam has more than sufficient tensile capacity to act as a horizontal tie in principle.	

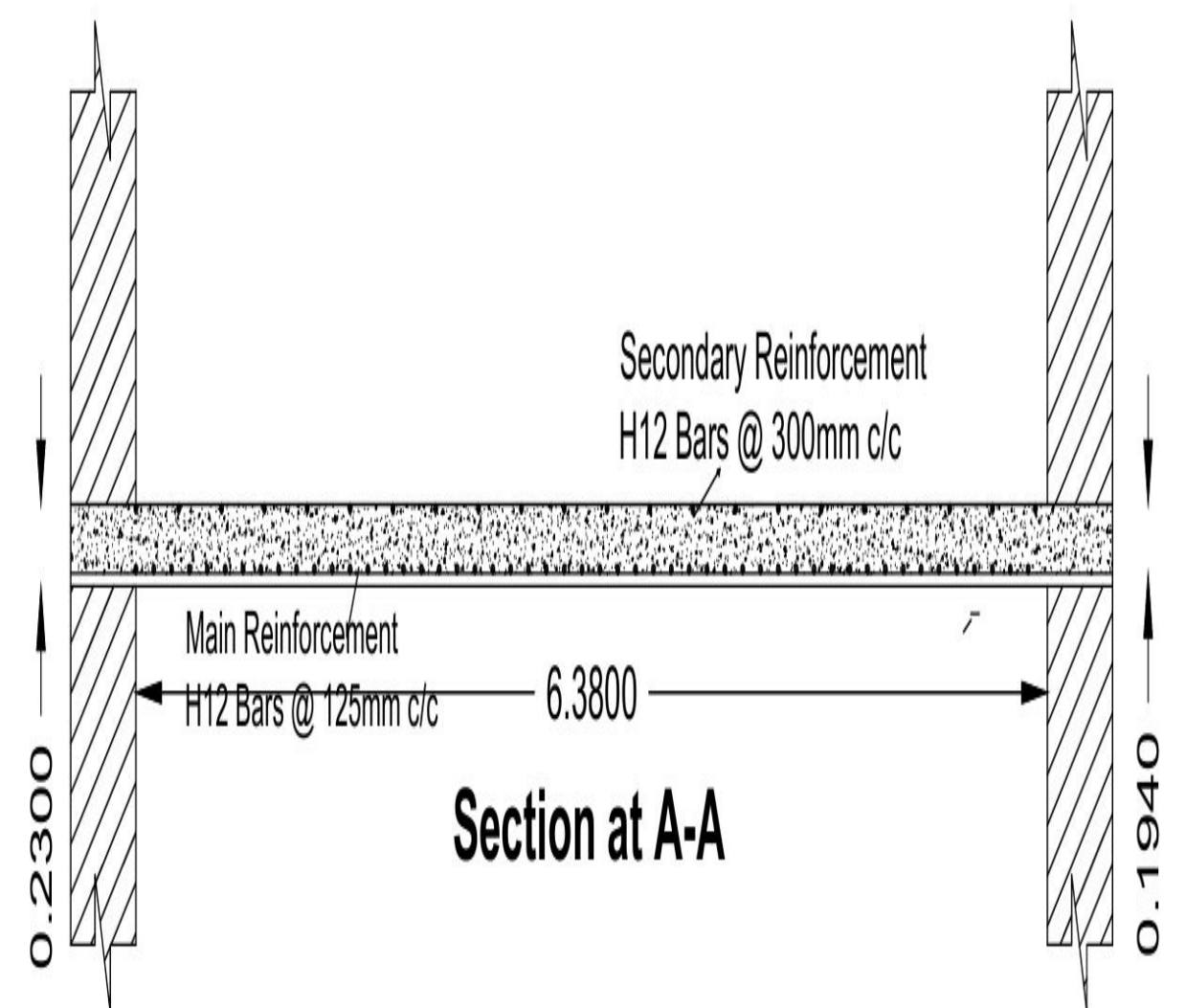
13

References	Calculation	Outcomes
BS EN 1993-1-2:2005 Cl. 2.4.2	Steel column fire check and slab From your column design, the governing buckling resistance is about the minor axis: $N_{Ed} = 188 \text{ kN}$ $N_{b,z,Rd} = 192 \text{ kN}$	
BS EN 1993-1-2:2005 Cl. 4.2.4	Load ratio: $\mu_0 = \frac{N_{Ed}}{N_{b,z,Rd}} = \frac{188}{192} = 0.98$	
BS EN 1993-1-2:2005 Cl. 4.2.4	Using the same critical temperature formula: $\theta_{a,cr} = 39.19 \ln \left(\frac{1}{0.9674 (0.98)^{0.333}} + 1 \right) + 482$ $\theta_{a,cr} \approx 511^\circ \text{C}$ For 60-minute fire resistance, the column satisfies provided their steel temperatures do not exceed 511 C For Slab The slab is adequate for 60-minute fire resistance due to sufficient concrete cover (30 mm) and slab depth (230 mm) in accordance with Eurocode 2 fire design (BS EN 1992-1-2).	
BS EN 1992-1-2:2004 EN 1992-1-2:2004 T.5.9	2) Column robustness check Column gross area from your section: $A = 29.2 \text{ cm}^2 = 2920 \text{ mm}^2$	
BS EN 1993-1-1:2005 Cl. 6.2.4	Using steel grade S235: $N_{pl,Rd} = \frac{A f_y}{\gamma_{M0}}$ $N_{pl,Rd} = \frac{2920 \times 235}{1.0} \times 10^{-3}$ $N_{pl,Rd} = 686 \text{ kN}$	
	Check against tie force: $686 \text{ kN} > 75 \text{ kN}$	
	Column robustness outcome The column has sufficient axial/tensile resistance to contribute as a vertical tie in principle.	

Structural Layout of Reinforced Concrete Slab (Plan and Section A-A)

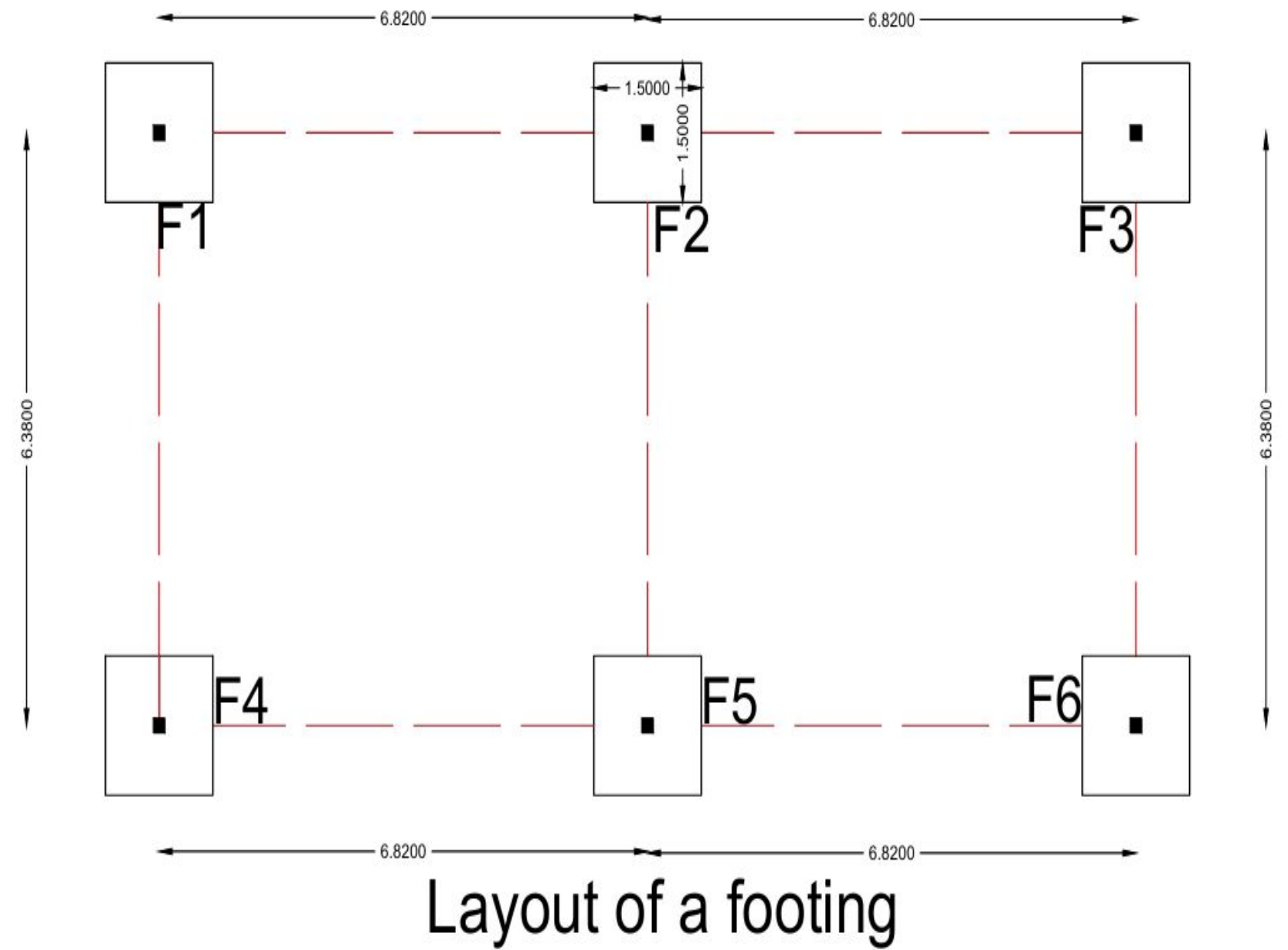
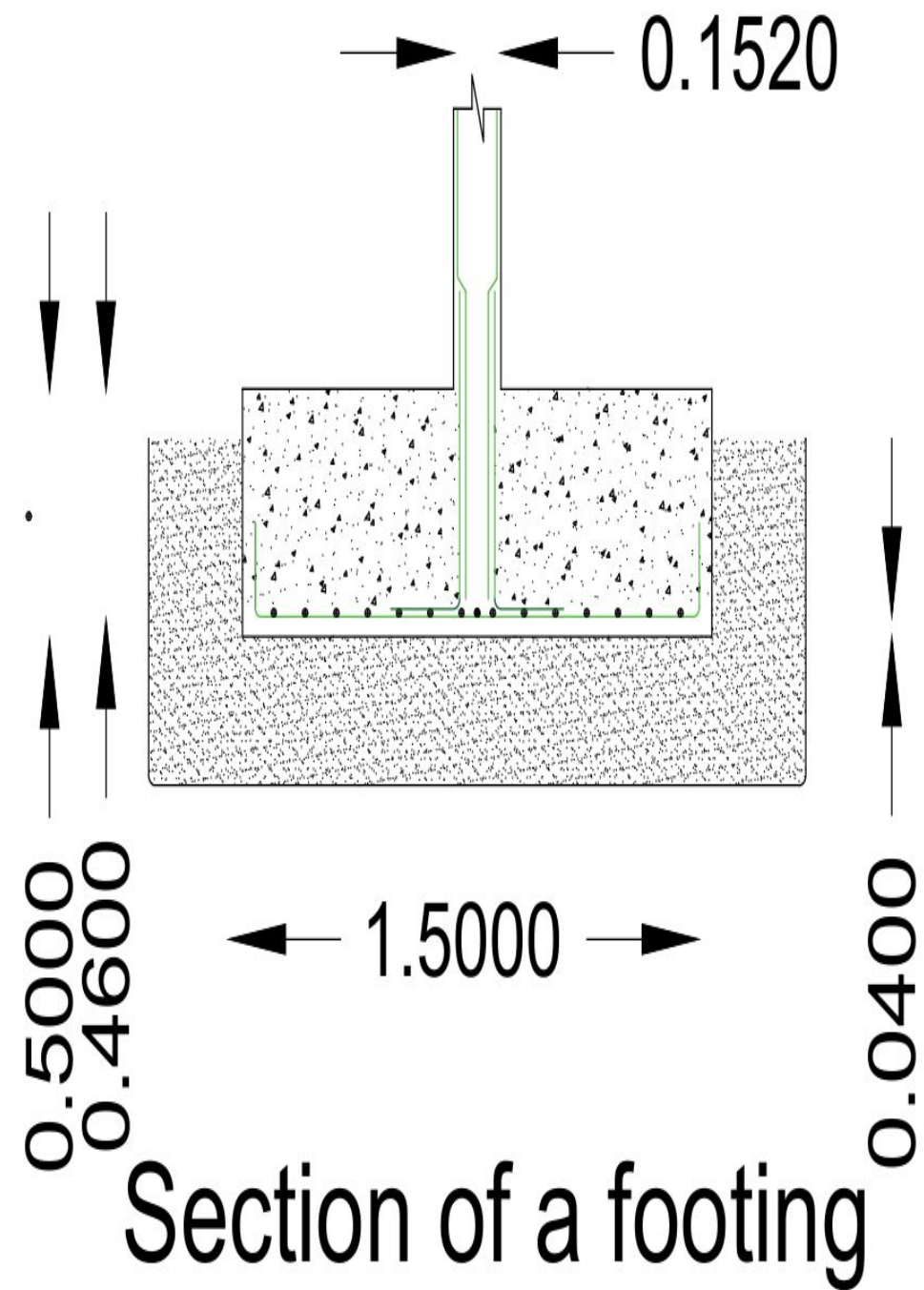


Plan View



Section at A-A

Foundation Layout and Section of Isolated Footings



Structural Appraisal for a Roehampton library existing structure

Summary (*Structural Surveys & Design Scotland Ltd, 2023*)

Roehampton Library, found at 2 Danebury Avenue, Roehampton, London, SW15 4HD, is a public library in the London Borough of Wandsworth. It plays a vital role in the community by offering access to books, study areas, and digital resources. The area around the library is mostly residential, featuring housing developments and community facilities. Notable landmarks nearby include the University of Roehampton and Richmond Park, which are among the largest green spaces in London. The library is well-connected locally, with bus routes and road access to Putney, Wandsworth Town, and Kingston Vale, making it easy for residents, students, and visitors to reach. Overall, its position in a mix of residential and educational surroundings strengthens its importance as a central place for learning.

Introduction

In relation to the project brief, a structural evaluation and condition check of the Roehampton Library, situated at 2 Danebury Avenue, London, has been completed. It is known that the building is being looked at for renovation and eco-friendly redevelopment, aiming to enhance its structural performance, energy efficiency, and overall usability while keeping the current structure where possible. This report has been requested to evaluate the present structural state of the building, suggest any repair work needed, and assess if the existing structure can be reused according to modern standards and design needs. The evaluation was based on a visual inspection of areas that were accessible and did not include any invasive checks on the foundations, walls, floors, or roof structure. There was enough access during the inspection to make a reasonable engineering assessment of the building's structural condition. (SSD Scotland, 2023)

Existing Building Description

Roehampton Library, situated at 2 Danebury Avenue in Wandsworth, is a one-story civic building from the 1960s that showcases post-war municipal architecture. The area around it is primarily residential and has pedestrian access. The building is thought to have a reinforced concrete frame along with prefabricated external panels. Its roof features concrete dome elements and a flat central section, which helps with efficient load distribution and allows for large internal spaces. Inside, the layout is mostly open-plan. In summary, the building represents the typical construction style of its era, but it falls short of today's standards for thermal efficiency and durability. (SSD Scotland, 2023)

External Survey Observations

During the external survey we observed that the outer walls are clearly showing signs of wear and tear from being exposed to the environment for a long time. There are noticeable stains and discoloration all over the surface, especially under projections and around openings, which suggests that water has been running off there consistently over time.

Dark streaks and uneven patches indicate that drainage is poor and moisture is building up. You can see specific areas where the surface is damaged and flaking, with some parts revealing the material underneath, which points to long-term weathering and possible moisture getting in. In some spots, you can find patch repairs and uneven finishes, hinting that previous maintenance efforts might not have worked well. (SSD Scotland, 2023)

Around where pipes go through the walls, there are stains and damp spots, which could mean there's a leak or that the detailing isn't done properly. At the lower levels, there's a lot of dirt buildup and general wear from exposure and use.

In summary, while there aren't any major structural cracks or failures visible, the external walls show several signs of aging, localized damage, moisture issues, and inconsistent maintenance that could impact their durability in the long run if not taken care



Fig:Surface Damage



Fig: pipe leaking and staining



Fig: Discoloration of wall

Internal Survey Observations

During the internal survey we observed that the internal inspection found multiple signs of damage and moisture problems inside the building. There are noticeable cracks and surface damage around the window openings, which could mean movement or wear overtime. Minor cracks can also be seen along joints and near fixings, hinting at localized stress or aging materials



Fig: Celling crack

At the ceiling level, especially around the dome junctions, there are obvious signs of water stains and discoloration. Brown patches and streaks suggest that water is getting in, probably from the roof above. In some spots, the ceiling surface shows signs of damage and possible peeling, indicating it has been exposed to moisture for a long time.



More cracks are visible on the internal wall surfaces, particularly near corners and around openings, which suggests minor structural or non-structural movement. There are small holes and exposed fixings, likely from past installations or maintenance work.

Fig: Celling damage and mould

In summary, while there are no major structural failures noted, the internal condition clearly shows signs of moisture content, cracking, and general deterioration. These problems highlight the need for further investigation and repair work to avoid long-term damage and ensure the building's durability (SSD Scotland, 2023)



Fig: Peeling of paint

Anticipated Remedial Works

1. Surface Damage (BRE 2007)

- Ø Remove any loose or damaged material.
- Ø Clean and treat any rust
- Ø Apply cement repair mortar.
- Ø Finish off with a matching render or pain

2. Pipe Leakage & Staining (BSI 2009)

- Ø Fix out the leaking pipe joint
- Ø Clean the stained spot
- Ø Put on waterproof sealant
- Ø Repaint using damp-resistant paint

3. Wall Discoloration (BRE 2007)

- Ø Identify the source of moisture
- Ø Clean the surface using a mild bleach solution
- Ø Apply a stain-blocking primer
- Ø Repaint

4. Ceiling Damage & Mould (WHO2009)

- Ø Fix the source of moisture
- Ø Clean the mould with a fungicidal wash
- Ø Replace any damaged plaster if necessary
- Ø Repaint using anti-mould paint

5. Ceiling Crack (BSI 2004)

- Ø Open the crack a bit (V-groove)
- Ø Use gypsum compound to fill it in
- Ø Sand it down and repaint

6. Paint Peeling (BSI 2009)

- Ø Remove any loose paint
- Ø Smooth the surface with sandpaper
- Ø Put on a coat of primer
- Ø Repaint using high-quality color

Summary and Conclusion

After reviewing the observations and assessments, we believe that the Roehampton Library is mostly structurally sound, with no immediate signs of significant structural failure noted during the visual check. However, there are several issues related to moisture entry, surface wear, and minor cracks throughout the building, suggesting that maintenance has been minimal in recent years.

While the main structural system, which likely includes a reinforced concrete frame and dome roof components, seems to be working fine, the state of both the inside and outside finishes shows that some repair work is needed to stop further damage. Specifically, moisture-related problems like staining, mould growth, and surface harm should be tackled as a priority to ensure durability and usability.(SSD Scotland, 2023)

Besides the repair work mentioned, more investigations are suggested, especially concerning the roof structure, drainage systems, and hidden structural parts. Since this assessment was based only on a non-intrusive visual inspection, it's possible that more defects could be discovered during detailed surveys or opening-up work.

The Roehampton Library is mostly in good shape structurally, and there are no significant structural problems seen. However, there are signs of moisture getting in, surface wear, and some small cracks that show it hasn't been maintained well. To stop more damage, repairs, especially for the moisture issues, are needed. It's also suggested to carry out more detailed investigations, as there could be other defects found during deeper inspections. In general, the building is fit for keeping and redeveloping, but it needs more evaluation.

Drainage Consideration

1. Location

SW15 4HD, London, United Kingdom

2. Description

The planned project is a library building that includes a main library area, a section for children, a reception area, and paved outdoor spaces around it. The site has non-porous roof surfaces and hard landscaping, which will lead to surface water running off when it rains.

To minimize runoff and enhance sustainability, a Sustainable Drainage System (Suds) strategy is implemented. This approach adheres to established guidelines such as the CIRIA Suds Manual (CIRIA, 2015) and similar case studies where elements like rain gardens and permeable paving are utilized to manage water efficiently (CIRIA, 2024).

3. Key Suds Elements Utilized

- ☐ Downspout planter
- ☐ Rain garden
- ☐ Swale
- ☐ Permeable paving
- ☐ Water butt (rainwater storage)

These elements are highly suggested for controlling surface water runoff and enhancing water quality (Environment Agency, 2022).

4. How It Works

Roof Drainage

Rainwater that falls on the roof is gathered by gutters and downpipes, which then channel it into downspout planters and water storage systems. This method helps to lessen the quick flow of water into the sewer and offers temporary storage (CIRIA, 2015).

Rain Garden System

Water from downpipes and paved surfaces is channeled into a rain garden situated in the green space. The rain garden:

1. Holds water for a short time
- 2.Allow infiltration to a ground
- 3.Cleans pollutants

This adheres to the Suds guidelines for managing and treating water (Environment Agency, 2022).

Swale (Boundary Drainage)

Excess water is directed towards swales along the site boundary.

The swale:

- ☐ Slows down water movement

☐ Helps in transporting water

☐ Lowers the chance of flooding
- Swales are often included in Sustainable Drainage Systems (Suds) to handle excess water safely (CIRIA, 2015).

5. Path of Surface Water Flow

- Roof → Downpipe → Planter / Water Butt
- Overflow → Rain Garden
- Excess → Swale
- Final → Infiltration or controlled discharge

This step-by-step method shows the best practices in Suds design (CIRIA, 2015; CIRIA, 2024)

6. Advantage

- Reduces surface water runoff
- Lowers flood risk
- Enhance water quality
- Allows for rainwater reuse
- Boosts biodiversity

These advantages align with the practical applications of Suds (CIRIA, 2024)

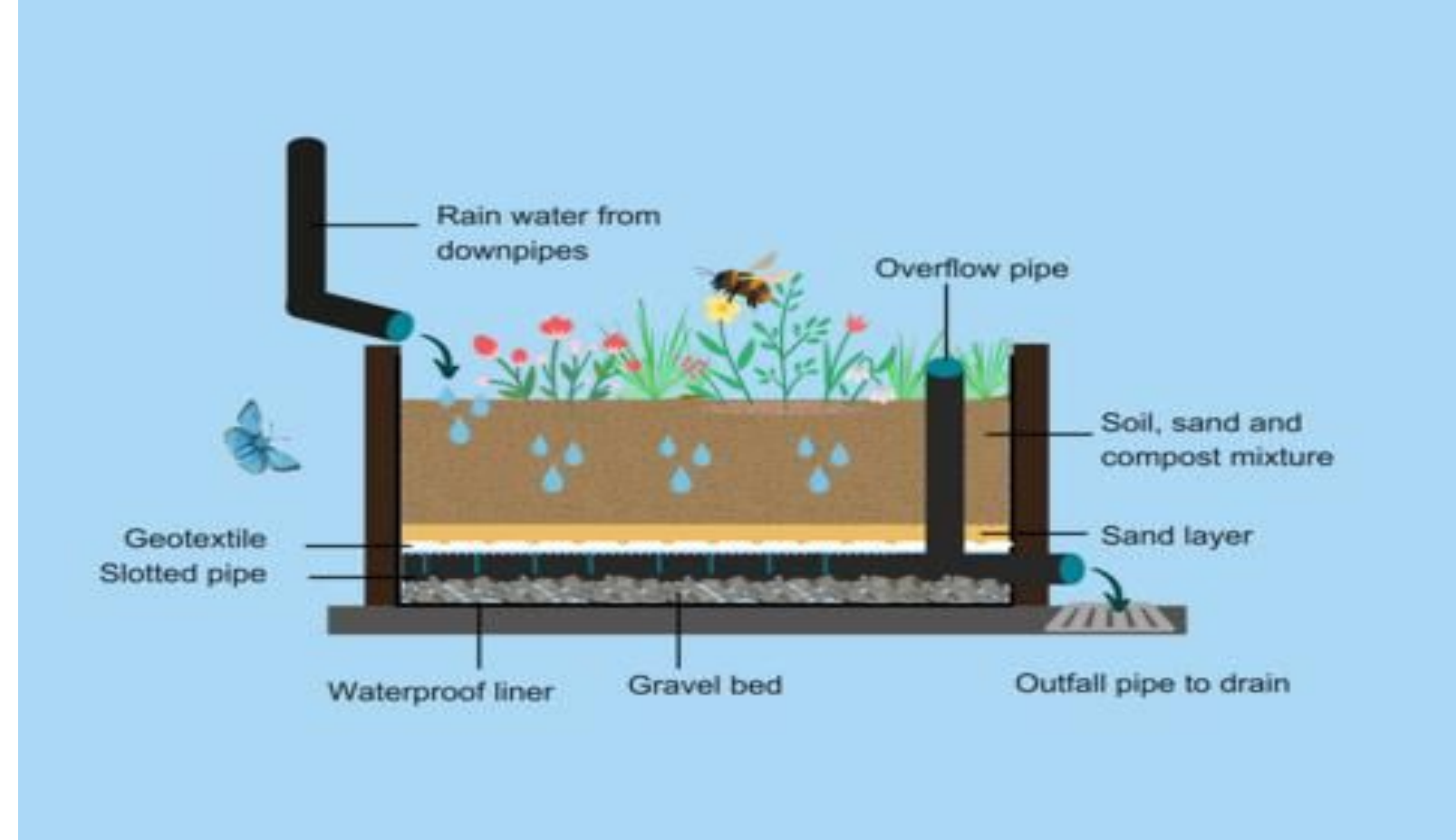
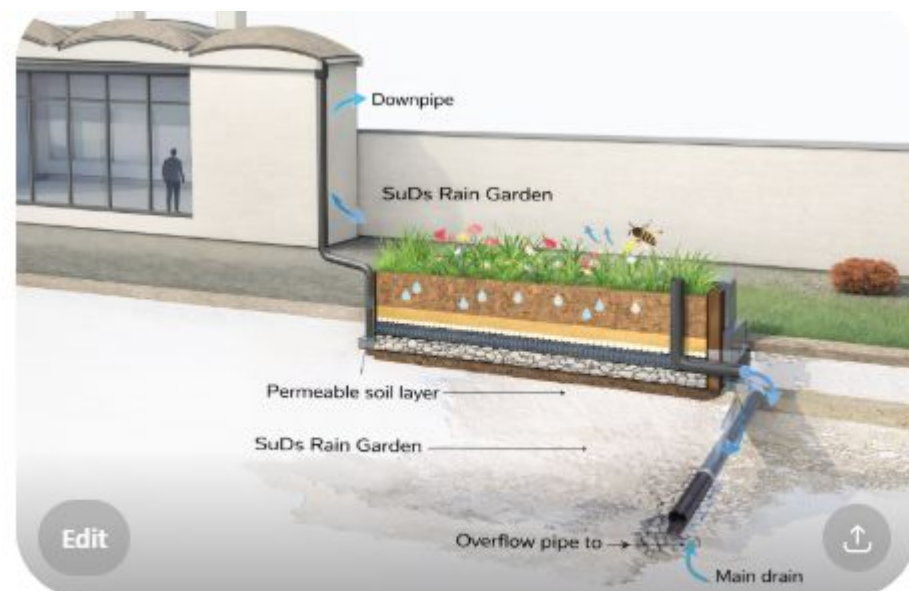


Fig:3D diagram showing surface water flow using Suds

7. Conclusion

The suggested drainage system follows the principles of Sustainable Drainage Systems (Suds) as outlined in established guidelines (CIRIA, 2015; Environment Agency, 2022). It includes features like permeable paving, rain gardens, swales, and rainwater storage to efficiently handle surface water runoff and enhance sustainability.

Fig:SuDS Rainwater Drainage Detail

Construction joint and Sustainability

Construction Joint

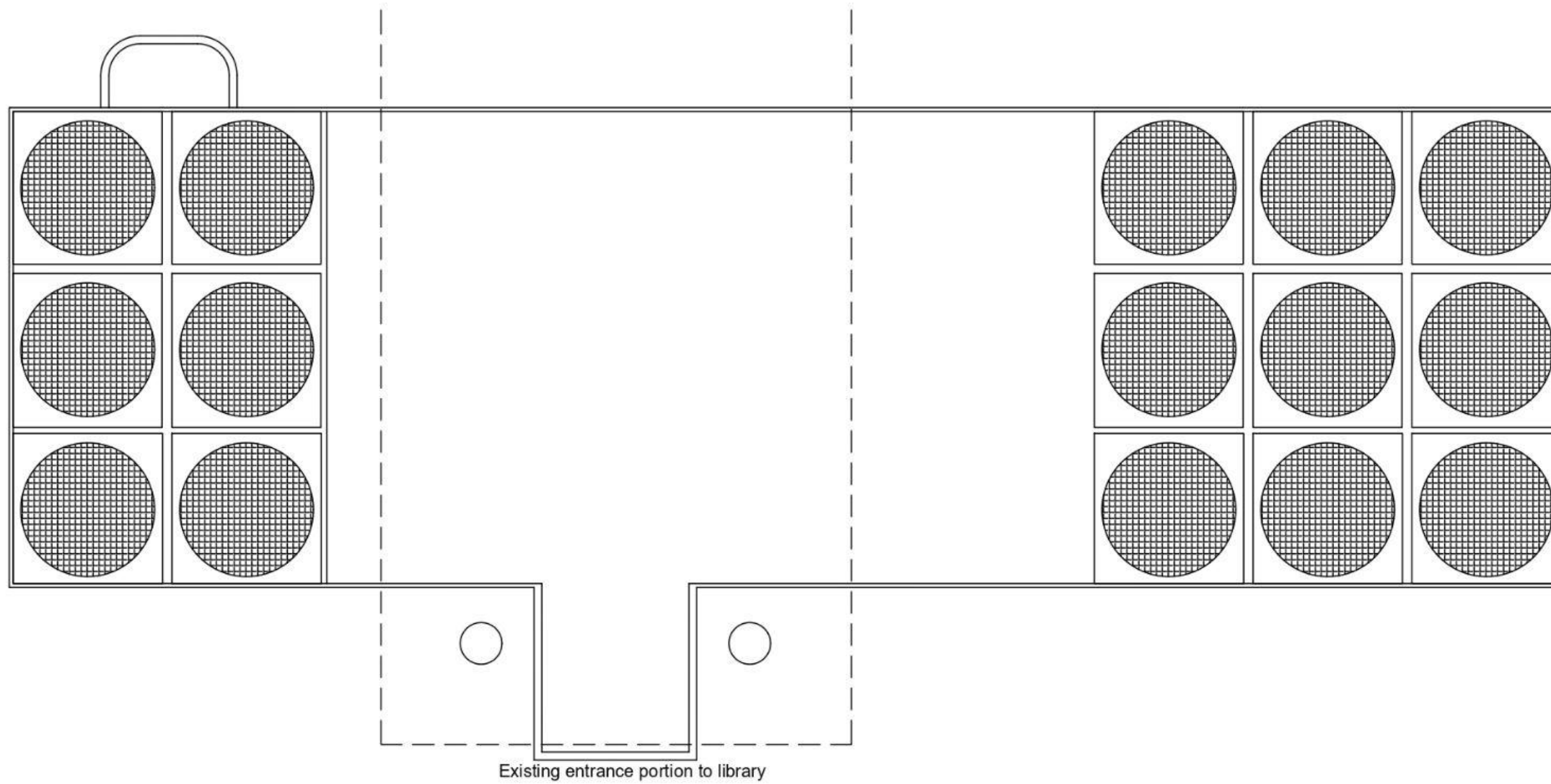
In this project, the new extension is built on separate foundations, while a moment-resisting connection is established between the new and existing structures to maintain continuity and effectively transfer bending moments. By using different foundations, the risk of uneven settling between the two structures is minimized, and the moment connection allows the structural elements to function together as a unified system, which enhances overall stiffness, stability, and load distribution, as mentioned in Reynolds's Reinforced Concrete Designer's Handbook. This method also improves structural performance when loads are applied. Attention is paid to deformation compatibility, including thermal effects and shrinkage, to avoid cracking that could result from differential settlement between the old and new foundations, ensuring that the connection remains effective over time (BS EN 1990, 2002).

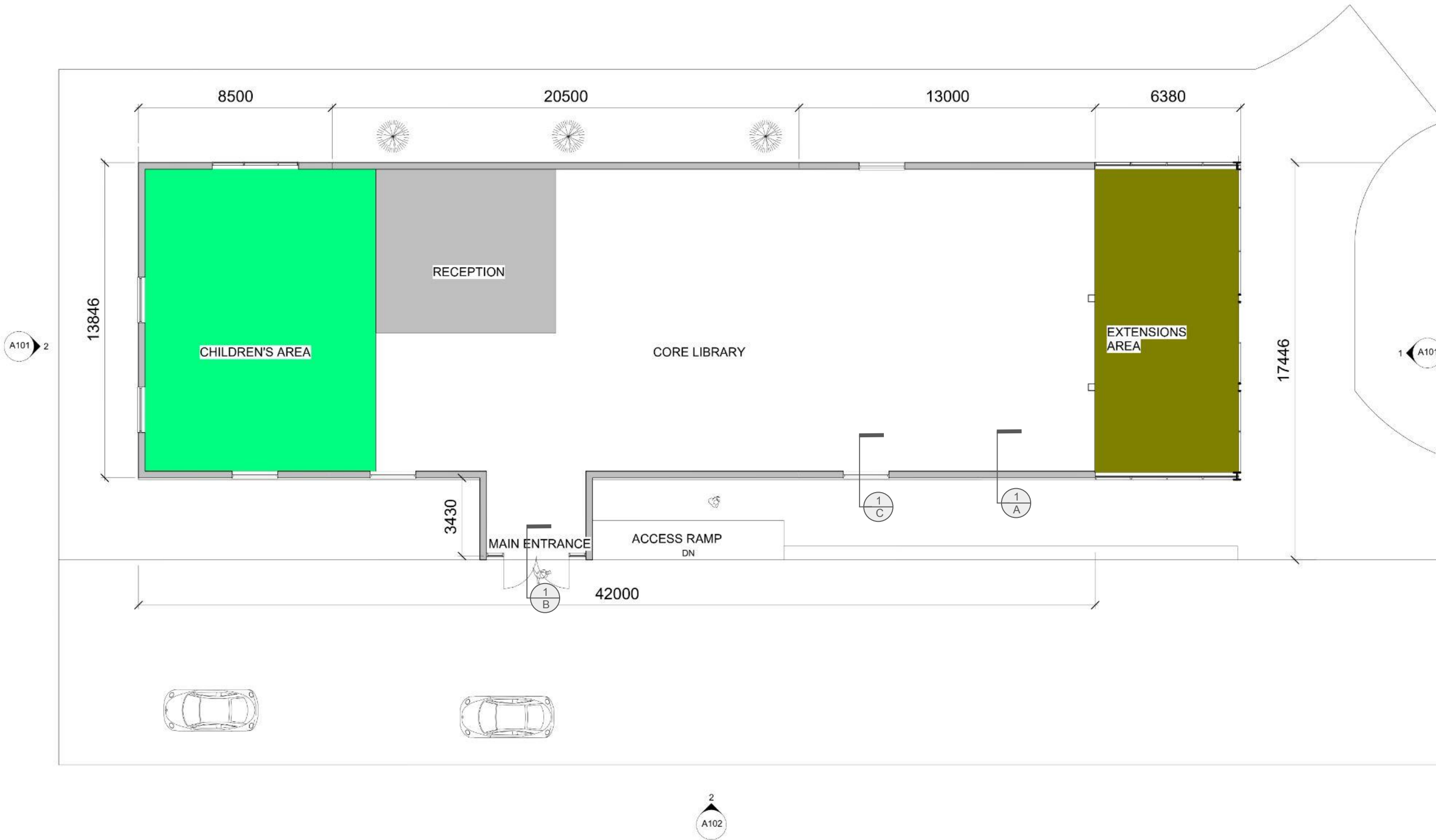
Sustainability

In this project, we focused on sustainability by using low-carbon concrete and low-carbon steel, which greatly cut down on embodied carbon compared to regular materials. Regular Portland cement concrete causes a lot of carbon emissions, releasing about 0.9 tonnes of CO₂ for every tonne of cement made, mainly because of clinker production and energy use (World Green Building Council, 2019). Likewise, traditional steel production is very carbon-heavy, producing around 1.8–2.0 tonnes of CO₂ for each tonne of steel (International Energy Agency, 2020). On the other hand, low-carbon concrete makes use of supplementary cementitious materials like fly ash or ground granulated blast furnace slag (GGBS), which helps to significantly lower the amount of cement needed and the emissions that come with it (Institution of Structural Engineers, 2020). Additionally, using low-carbon or recycled steel helps to further decrease embodied carbon by lessening the need for primary steel production (World Steel Association, 2021). As a result, the design we proposed helps to lower the overall carbon footprint and promotes sustainable building practices.

Drawings

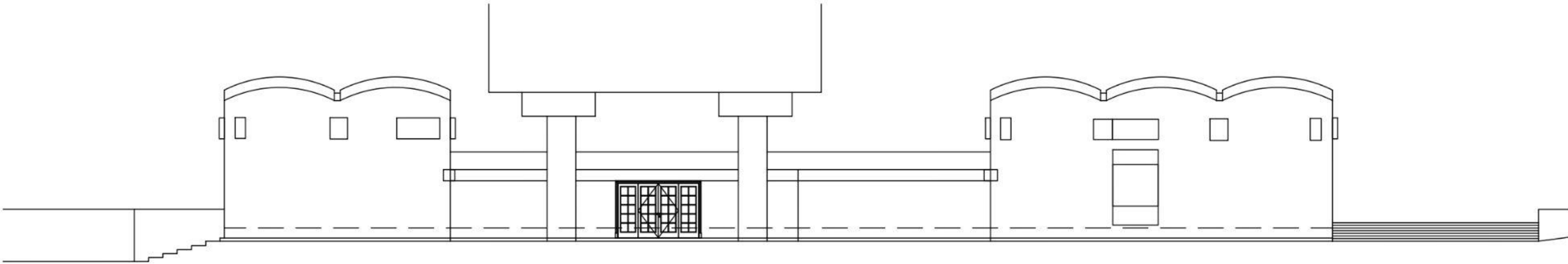
Roof Plan (Existing) 1:200





ELEVATIONS

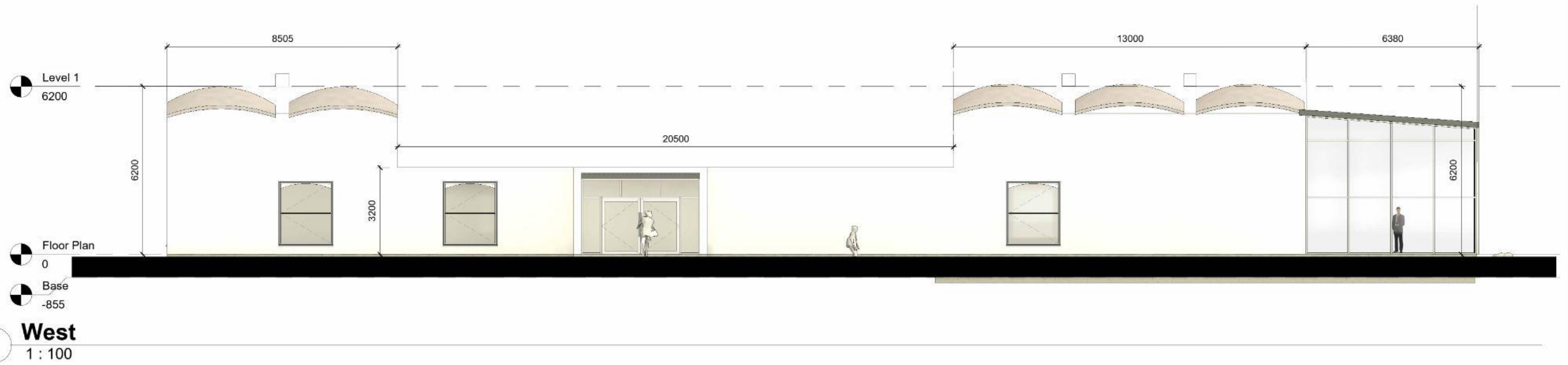
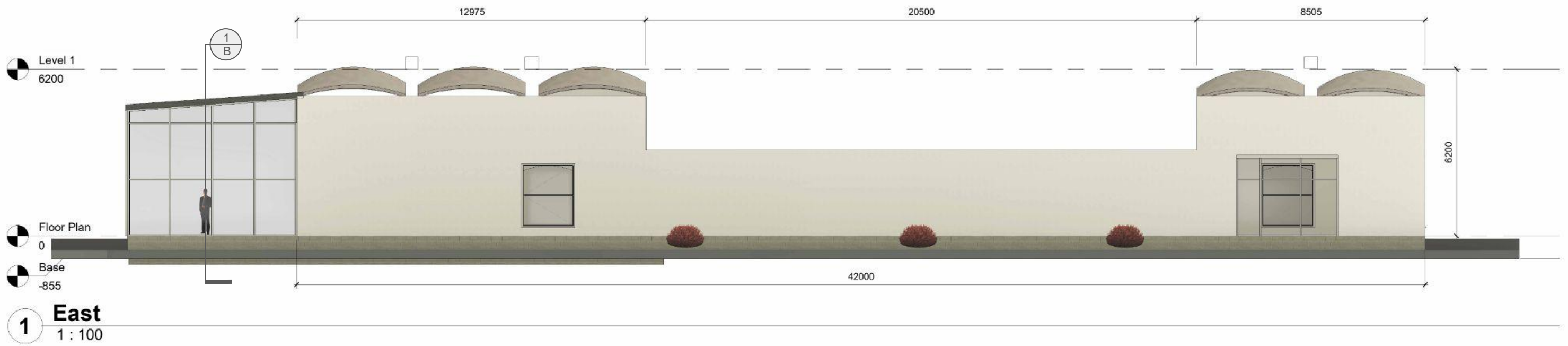
Roehampton Library Existing Elevation
Scale - 1:150

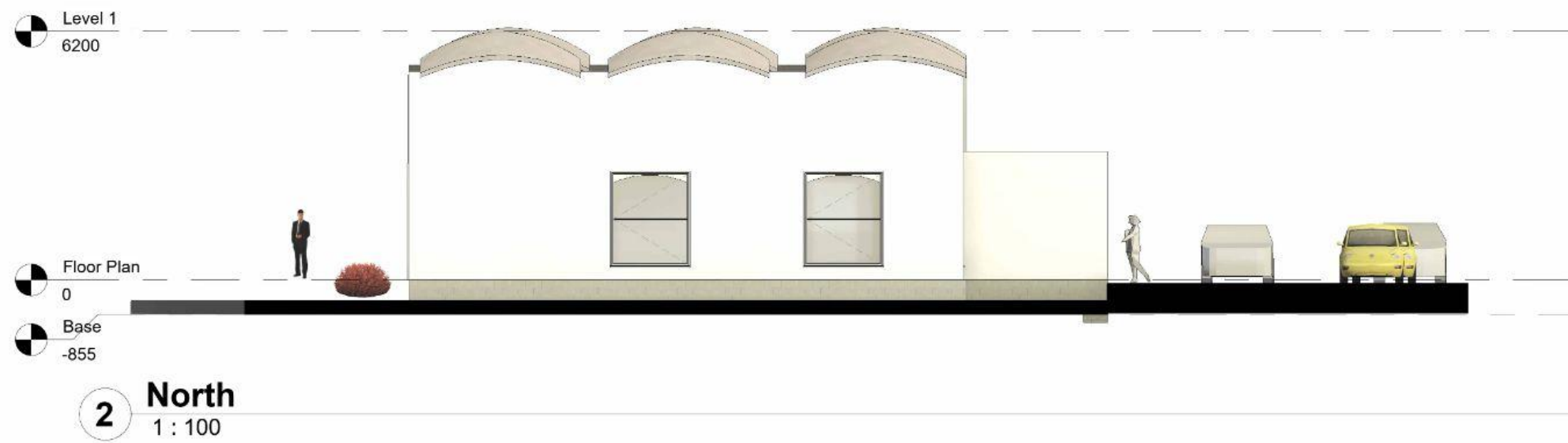


South Elevation (Existing)



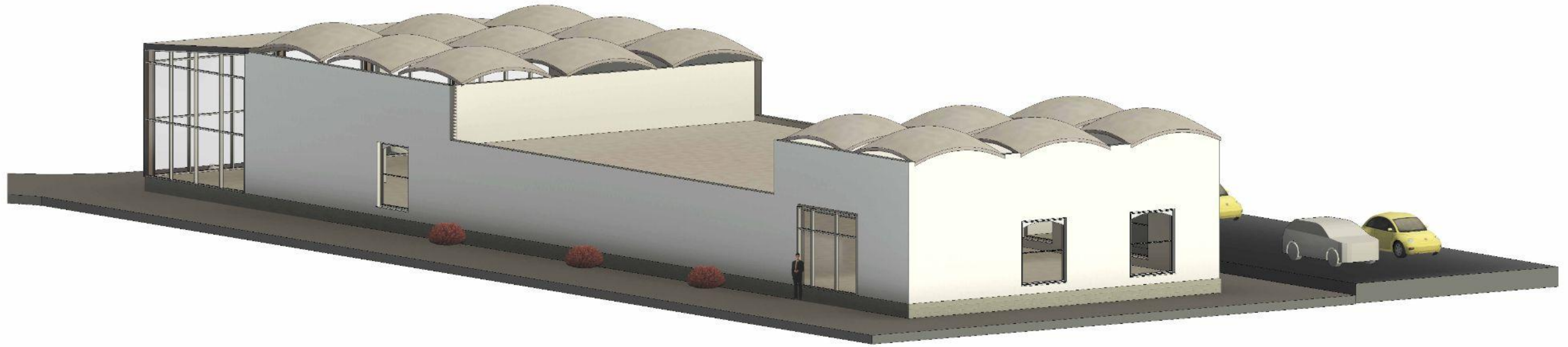
East Elevation (Existing)





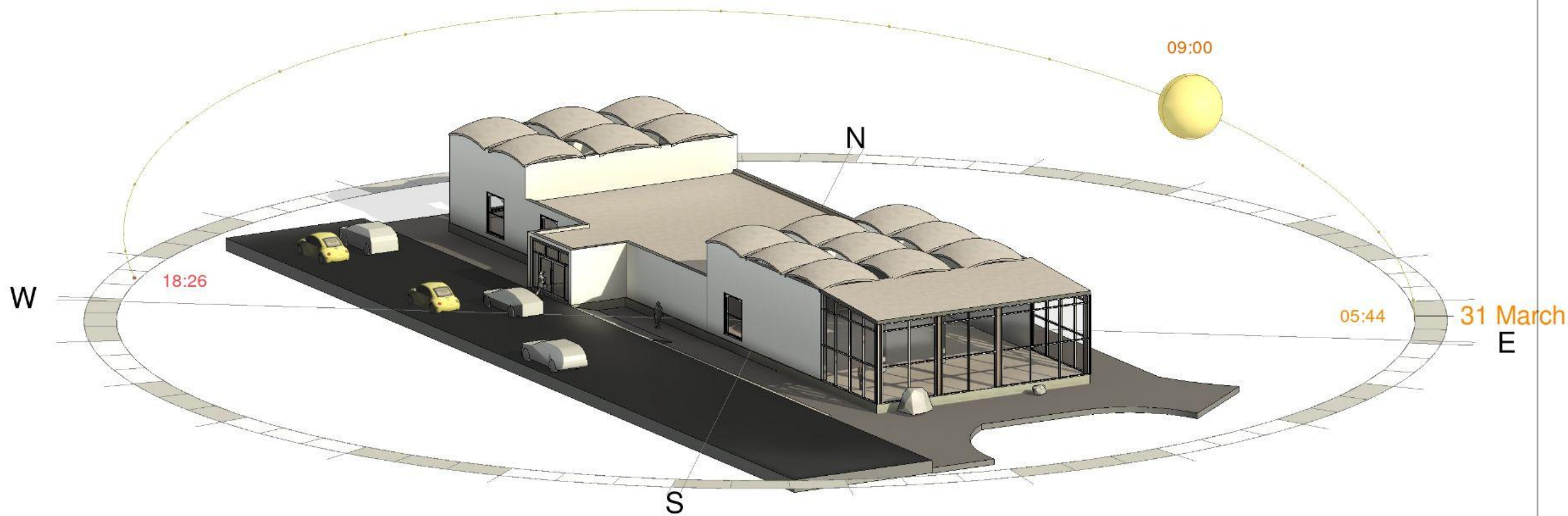
3D VIEWS





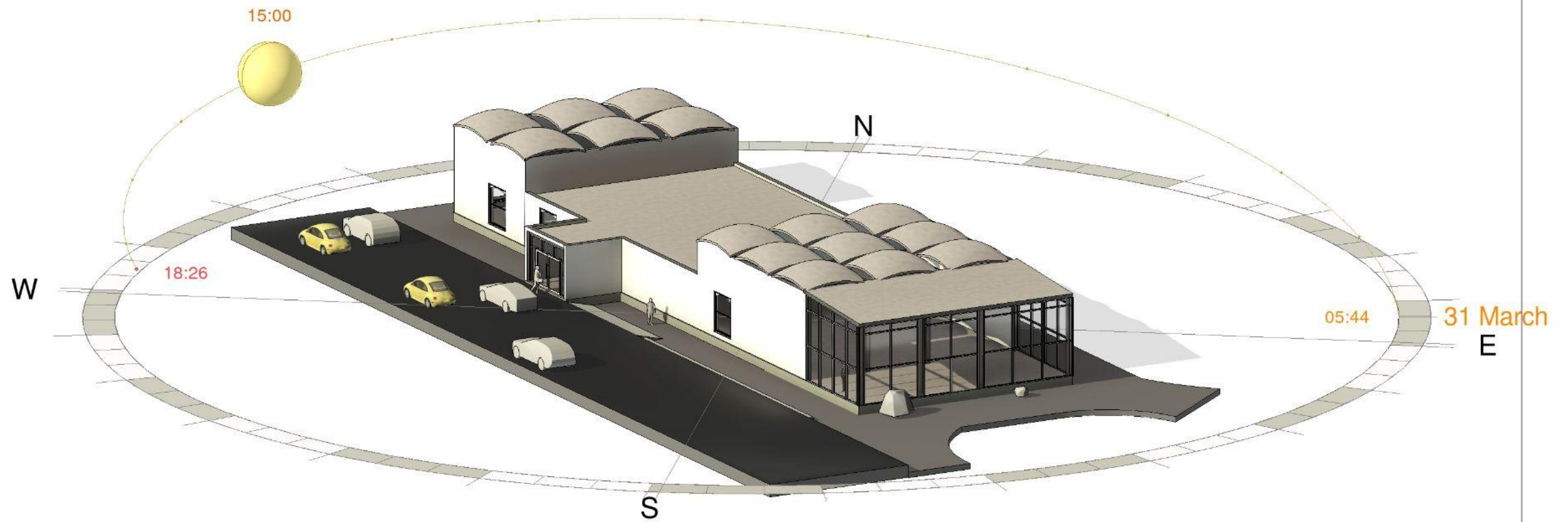






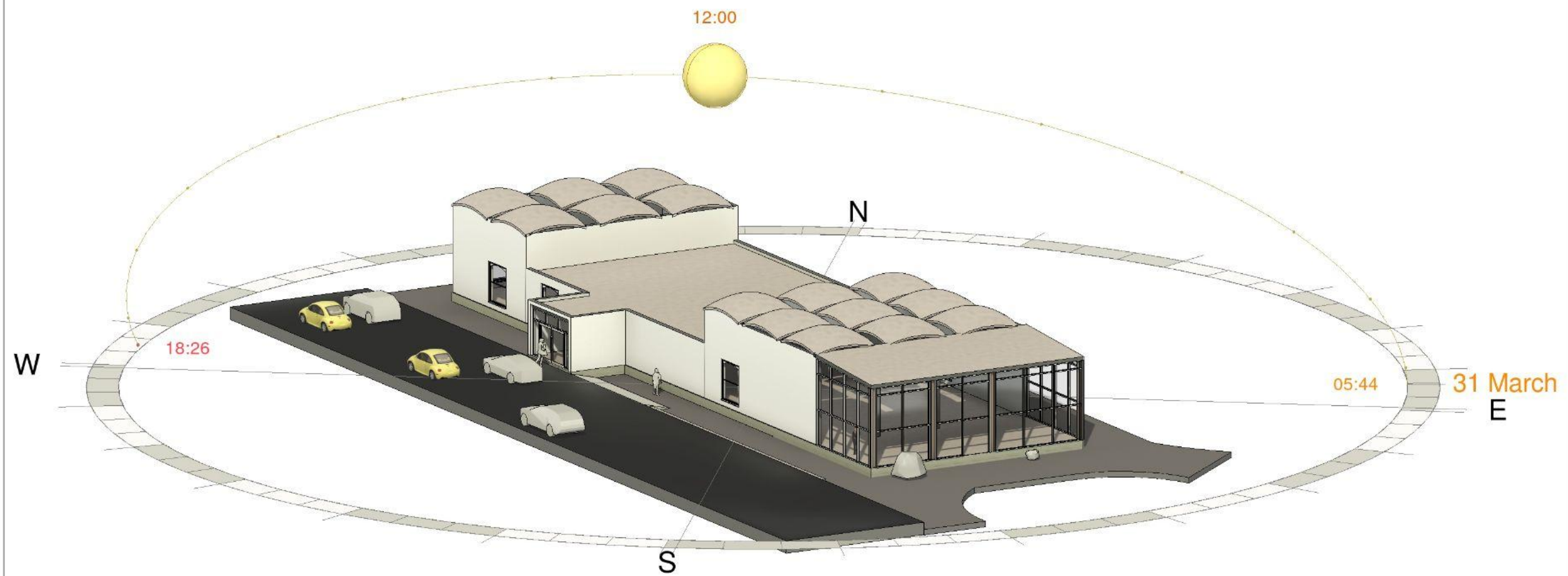
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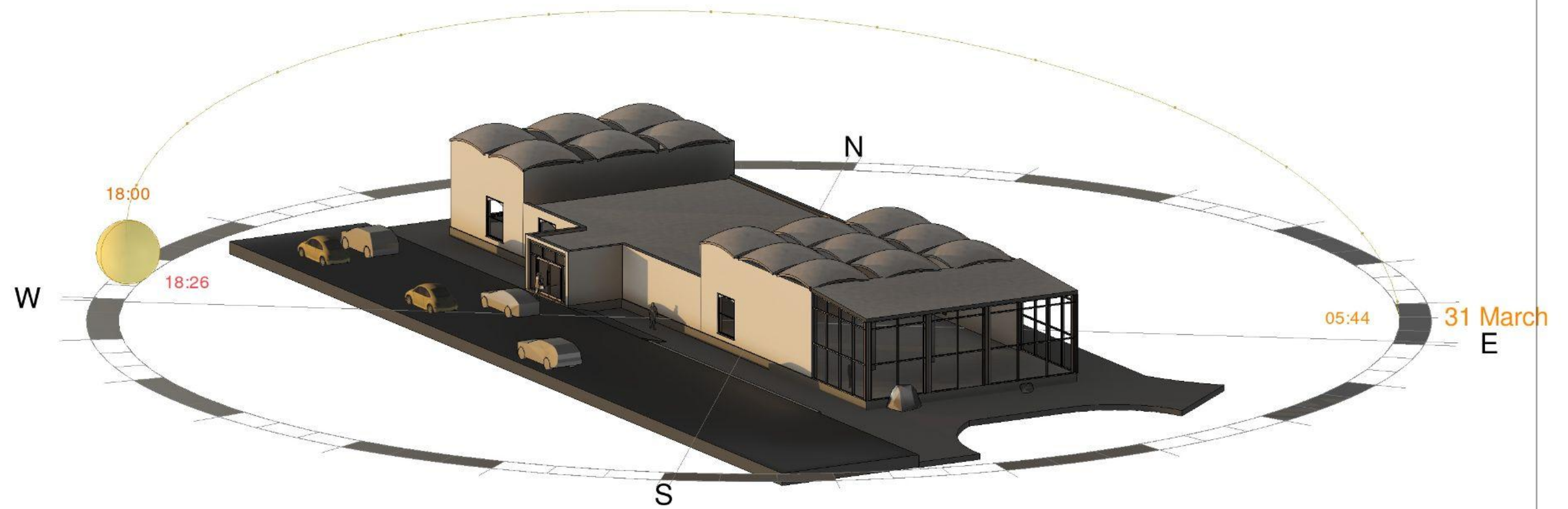
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1 Sun Path Analysis - 1200hrs

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1 Sun Path Analysis - 1800hrs

11/04/2026 16:16:46

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